MULTIMODAL COMPUTATIONAL INTELLIGENCE FOR ENHANCING CAREGIVER PERFORMANCE AND CARE DELIVERY

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ABSTRACT

Caregivers in care homes record Activities of Daily Living (ADL) such as meals, medication, hygiene, mobility and symptoms, yet in many Sri Lankan facilities these records are still handwritten. Field visits to an elder care home showed that handwritten records are hard to maintain and that observations are sometimes missed. Existing speech and documentation tools mainly produce transcripts or free-text notes; they do not turn caregiver speech into structured ADL records, resolve spoken self-corrections, or flag safety-relevant observations. This research developed SCRIBE, the documentation component of the CareSense framework, which converts caregiver voice logs into structured and safety-aware ADL records. The pipeline adapts Whisper-small to accented English with Low-Rank Adaptation (LoRA), removes fillers and self-corrections with a fine-tuned BART-base model, corrects grammar while protecting medication and food names, and extracts a 19-field ADL record with an alert flag using a fine-tuned T5-small model. Training data combined public speech and disfluency corpora with a de-identified dataset of 2,325 ADL records built from the care home's handwritten records and caregiver input. LoRA adaptation reduced the word error rate from 26.13% to 15.57%. T5-small detected all 54 alert cases in its held-out test split, but alert recall fell to 82.0% on a separate 200-case safety set; deriving the alert from the extracted fields raised recall to 93.3% with no false alarms. Audio chunking and sentence segmentation allowed long, multi-observation recordings to be processed, raising event recall on 100 multi-observation test paragraphs from 41.2% to 100%, and a 51-second recording was processed in 12.2 seconds on a GPU. The work contributes an accent-adapted, safety-aware ADL documentation pipeline, a de-identified ADL dataset and a safety-focused evaluation method. Testing with real caregiver speech in the care home is the main next step.

Keywords: Activities of Daily Living, caregiver documentation, speech recognition, disfluency removal, clinical information extraction

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADL	Activities of Daily Living
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
BART	Bidirectional and Auto-Regressive Transformers
BLEU	Bilingual Evaluation Understudy
BP	Blood Pressure
CER	Character Error Rate
CPU	Central Processing Unit
EHR	Electronic Health Record
EM	Exact Match
FN	False Negative
FP	False Positive
FR	Functional Requirement
GPU	Graphics Processing Unit
gTTS	Google Text-to-Speech (Python library)
IADL	Instrumental Activities of Daily Living
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NFR	Non-Functional Requirement
NLP	Natural Language Processing
PDPA	Personal Data Protection Act
PEFT	Parameter-Efficient Fine-Tuning
ROUGE	Recall-Oriented Understudy for Gisting Evaluation

Abbreviation	Meaning
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
T5	Text-to-Text Transfer Transformer
TN	True Negative
TP	True Positive
TTS	Text-to-Speech
WER	Word Error Rate

1 INTRODUCTION

1.1 Overall Project Context

Care homes for older adults depend on caregivers who look after care recipients throughout the day and night. Their work includes helping with meals, giving medication, bathing, moving care recipients safely, watching for changes in health and passing information to the next shift. At the same time, these facilities often have limited staff and limited technology. The group project, CareSense - Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery, was designed to support caregivers and improve patient safety in this setting by bringing together several artificial intelligence (AI) components that each address a different practical problem.

CareSense consists of four individual components, PULSE, SENTRY, SCRIBE and GLOSS, shown in Figure 1.1. They are mentioned briefly here only to place this work in context. PULSE, the first component, monitors caregiver biometrics and work logs to detect workforce deterioration and suggests fair shift adjustments to prevent burnout. SENTRY, the second component, analyses anonymised skeletal movement at the edge to detect instability and send real-time fall-risk alerts. GLOSS, the fourth component, uses computer vision to recognise sign language in real time and delivers customised lessons and feedback to caregivers.

SCRIBE, the third component, is the subject of this thesis. Its purpose is to reduce the burden of handwritten documentation in care homes by allowing caregivers to record their observations by voice. SCRIBE is a context-aware intelligent system that converts caregiver voice logs into structured documentation of Activities of Daily Living (ADL). A caregiver speaks an observation, such as a meal, a medication, a bath or a new symptom, into a web application. SCRIBE then transcribes the speech with an accent-adapted speech recognition model, removes fillers and resolves spoken self-corrections, corrects the grammar without changing medication or food names, extracts a structured ADL record, and decides whether the observation needs the supervisor's attention. Finally, it produces a written report and a spoken handover summary for the next shift. In this way, SCRIBE provides the automated care

documentation part of CareSense, turning everyday caregiver speech into records that can be searched, reviewed and handed over.

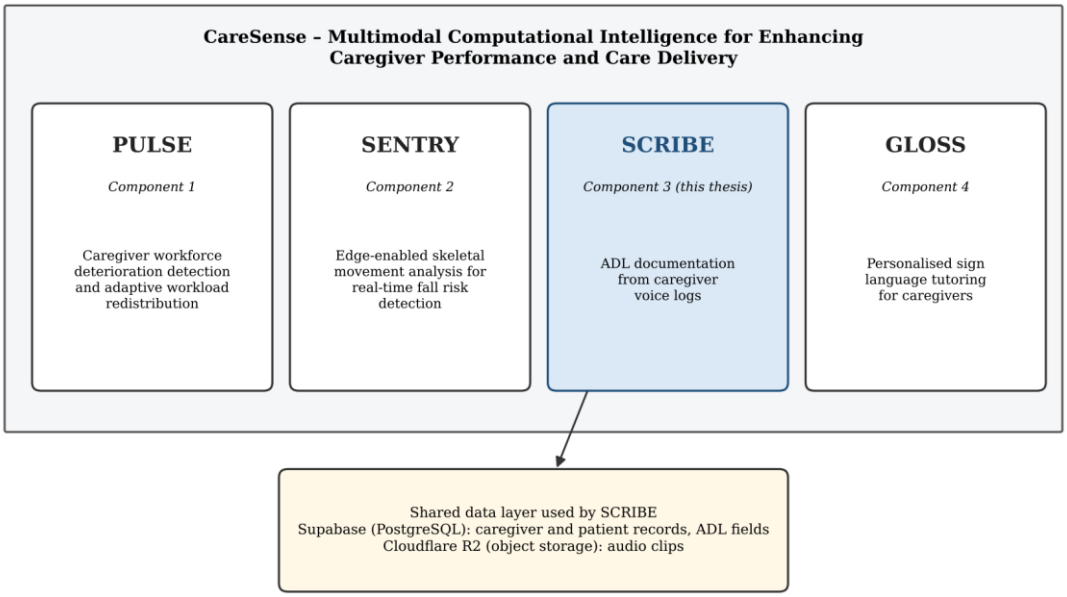


Figure 1.1: CareSense framework with its four components (PULSE, SENTRY, SCRIBE and GLOSS) and the position of SCRIBE

Within CareSense, SCRIBE stores structured information (caregiver and patient records and the extracted ADL fields) in a shared Supabase PostgreSQL database and stores unstructured data (audio clips) in Cloudflare R2 object storage. This allows the documentation produced by SCRIBE to be used by the wider CareSense platform rather than staying in a separate tool.

1.2 Individual Component Context

Activities of Daily Living are the basic self-care tasks that a person performs every day. The Katz index groups them into bathing, dressing, toileting, transferring, continence and feeding [1], while instrumental activities such as managing medication and preparing food extend this list [2]. In a care home, recording these activities is not optional paperwork. The records show what care was given, whether a care recipient ate and drank enough, whether medication was taken, and whether any new symptom appeared. Clinical documentation research also shows that both structured data and narrative notes are needed: structured records support reporting and decision systems, while narrative text keeps the detail and context that clinicians rely on [3].

At the Love & Elder Care Home, where the field work for this research was carried out, each care recipient has a paper file and caregivers write observations, medication and vital-sign readings by hand. Caregivers explained during the visits that maintaining these handwritten records is difficult and that some observations are missed. A second care home that was visited described the same problem. SCRIBE was designed for this situation. Instead of writing, a caregiver speaks a short observation into a web application. The system transcribes the speech, removes fillers and self-corrections, corrects the grammar, extracts a structured ADL record and decides whether the observation needs an alert for the supervisor. It then produces a written report and a spoken handover summary.



Figure 1.2: Common area of the Love & Elder Care Home (faces obscured)

1.3 Problem Statement

Documentation takes a large share of care staff time and interrupts direct care. Reviews of clinical documentation have studied how much time hospital staff spend on documentation and how electronic record systems change this burden [4]. The burden matters for patients as well as staff. When records are late, incomplete or unclear, the next caregiver may not understand the care recipient's recent condition. Studies of information transfer at hospital discharge found that missing or delayed summaries

affected continuity of care and patient safety [5]. Although discharge summaries are different from daily ADL logs, the same principle applies to shift handover in a care home: if an observation is not written down clearly, the next caregiver cannot act on it.

The field visits confirmed this problem in a local setting. Records were handwritten in individual patient files, the format varied between caregivers, and caregivers reported that observations were sometimes missed. Speech is a natural alternative, because caregivers can describe what they did while their hands are busy. However, real caregiver speech is not clean text. It contains fillers such as "uh" and "um", repeated words, and self-corrections such as "I gave her two, no wait, one tablet". It is also spoken with South Asian accents that general speech recognition systems handle less well than native English accents [6]. A transcript alone also does not give a structured record or tell a supervisor which observation is urgent.

Problem statement. Caregivers in care homes need a way to document ADL quickly and accurately without handwriting, but existing tools do not reliably convert accented, disfluent caregiver speech into structured ADL records that capture the corrected meaning and identify safety-relevant observations for supervisors.

1.4 Research Gap

Several lines of research address parts of this problem, but none of them covers the full path from caregiver speech to a structured, safety-aware ADL record. A systematic review of natural language processing (NLP) for ADL information found 22 studies, describing 20 systems, that extract ADL information from written electronic health record (EHR) notes [7]. The review reported that NLP is promising but limited by restricted access to EHR data, inadequate documentation, a lack of granularity and small datasets. These systems work on written clinical notes, not on spoken caregiver logs.

Digital scribes that use ambient listening and generative AI have been studied mainly for physician documentation. A rapid review found that they reduced self-reported documentation time, but it also reported instances of generated content that was not supported by the conversation, which means notes still need careful review [8]. These

tools produce free-text clinical notes rather than structured ADL fields, and they are designed for clinician consultations rather than caregiver observations. Disfluency correction has been studied with large general-purpose corpora such as DISCO [9] and DisfluencySpeech [10], but these corpora do not contain caregiving language such as medication doses or local food names. Accented speech recognition benchmarks such as Svarah show that Indian-accented English remains challenging for existing systems [6], and no public dataset exists for caregiver ADL extraction.

The gap addressed in this research is therefore an integrated, lightweight pipeline that (1) adapts speech recognition to accented caregiver speech, (2) resolves fillers and spoken self-corrections so that only the intended meaning is kept, (3) extracts category-specific ADL fields into a structured record, and (4) evaluates and improves the detection of safety-relevant observations, using a dataset grounded in real care-home records. This gap is analysed in more detail in Section 2.5.

1.5 Aim and Objectives

Main objective. The main objective of this research is to design, develop and evaluate a context-aware intelligent system that can automatically convert caregiver voice logs into structured Activities of Daily Living documentation, with the purpose of improving documentation accuracy, reducing manual workload, and supporting more effective care reporting and shift handover in caregiving environments.

Specific objectives. The specific objectives, as defined in the research proposal, are:

1. To examine the current methods used by caregivers to record ADL information and to identify the main challenges associated with manual documentation, including delayed recording, incomplete notes, inconsistency and increased administrative burden.
2. To develop a speech recognition component that can accurately convert caregiver voice logs into text by recognising care-related vocabulary, routine caregiving expressions and naturally spoken language commonly used in healthcare and caregiving settings.
3. To design and implement an auto-editing component that can identify and handle fillers, hesitations, repetitions and spoken self-corrections in caregiver speech,

so that only the final intended meaning is preserved for clear and reliable documentation.

4. To develop a domain-specific natural language processing module that can automatically extract and structure key ADL-related information from caregiver voice logs, including medication details, food and fluid intake, mobility assistance, hygiene activities, symptoms, time references and emotional observations.
5. To generate structured ADL records and concise shift handover summaries from the extracted information, available both as a text report and in audio format, and to evaluate the proposed system in terms of accuracy, completeness, usability and its ability to support continuity and quality of patient care.

1.6 Research Questions

The objectives were translated into four research questions that guided the experiments.

- **RQ1.** How much can a lightweight speech recognition model (Whisper-small) be improved for accented English through parameter-efficient adaptation, and how do the amount of training data and the LoRA rank affect accuracy?
- **RQ2.** Can a fine-tuned sequence-to-sequence model remove fillers, repetitions and spoken self-corrections more reliably than a rule-based approach, and which architecture handles safety-relevant corrections such as medication doses better?
- **RQ3.** Can a small sequence-to-sequence model extract a structured 19-field ADL record and an alert flag from a caregiver observation, and how reliably are safety-relevant alerts detected for phrasings that were not seen during training?
- **RQ4.** How can the pipeline process long recordings that contain several observations, and how quickly can it process a realistic recording?

1.7 Scope, Limitations and Assumptions

The scope of this component is the documentation of caregiver observations in English, as spoken by caregivers with South Asian accents. The ADL schema covers eight observation categories: meals, medication, hygiene, mobility, symptoms, fluid intake, nurse checks of vital signs, and family visits. The system is a documentation

and alerting aid. It does not diagnose, it does not replace the judgement of nurses or supervisors, and emergencies must still be reported directly by staff.

Several limitations and assumptions apply. No audio was recorded at the care home; the care-home data consisted of handwritten records and information from conversations with caregivers, and the speech model was adapted using the public Svarah corpus of Indian-accented English rather than Sri Lankan caregiver speech. The ADL dataset was generated with the help of a large language model (LLM) from de-identified care-home information, so its sentences are more regular than real caregiver speech. The extraction model assumes that each sentence describes one observation; long recordings are therefore split into sentences before extraction. Sinhala, Tamil and code-mixed speech are outside the scope of this work. At the time of writing, the ethics application submitted to SLIIT was still awaiting a decision, so all work used de-identified information only (Section 3.6).

1.8 Individual Contribution

All work described in this thesis was carried out by the author as an individual component of the CareSense group project. The individual contributions are:

- Field study at the Love & Elder Care Home: collection of handwritten records and conversations with caregivers, and manual de-identification and structuring of this information.
- Construction of a de-identified ADL dataset of 2,325 records with a 19-field schema and alert labels, and of a 200-case safety-focused test set.
- LoRA adaptation of Whisper-small with data-efficiency and rank ablations, reducing the word error rate (WER) from 26.13% to 15.57%.
- Fine-tuning and comparison of BART-base and T5-small for disfluency removal and for ADL extraction, with model selection weighted towards safety-relevant cases.
- A safe grammar-correction step with a protected medication and food vocabulary, and a field-based alert rule layer.

- Long-recording support through audio chunking and sentence segmentation, added in response to the second progress presentation (PP2) feedback, and end-to-end timing on GPU and CPU.
- Template-based written reports and spoken handover summaries, and the SCRIBE web application and backend integrated with the shared CareSense data layer.

1.9 Thesis Structure

Chapter 2 reviews the literature and existing solutions and identifies the research gap. Chapter 3 describes the research design, the data, the experimental procedure, the evaluation metrics and the ethical considerations. Chapter 4 presents the requirements and the design of SCRIBE, and Chapter 5 describes its implementation, integration and testing. Chapter 6 reports the results, and Chapter 7 discusses them, including limitations. Chapter 8 considers commercialisation and sustainability, and Chapter 9 concludes the thesis and outlines future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The literature review was organised around the stages of the SCRIBE pipeline rather than as a formal systematic review. Five themes were covered: (1) ADL and clinical documentation in care settings, (2) speech recognition for accented English, (3) disfluency and self-correction handling, (4) extraction of structured information from clinical and care text, and (5) AI-assisted documentation tools. Sources were taken from peer-reviewed venues in speech and language processing (the ACL Anthology and the ISCA archive), machine learning conferences, health informatics journals, and the documentation of the open-source tools used in the implementation. Preference was given to work that released datasets or models that could be reused, and to reviews that summarised the state of a field. The review informed both the design choices in Chapter 4 and the comparison in Chapter 7.

2.2 Domain and Theoretical Foundations

2.2.1 Activities of daily living and care documentation

The concept of ADL comes from geriatric assessment. Katz and colleagues defined an index of basic activities (bathing, dressing, going to the toilet, transferring, continence and feeding) to measure a person's independence [1]. Lawton and Brody added instrumental activities, such as taking medication and preparing food, which require more complex abilities [2]. In care homes, these activities are recorded as part of daily care so that staff can see whether a care recipient is eating, drinking, moving and taking medication as expected, and can notice changes early.

Clinical documentation has two forms that serve different purposes. Structured data, such as coded fields, can be searched, counted and used by decision-support systems, while flexible narrative text records context and nuance that fixed fields cannot capture [3]. A documentation system for caregivers therefore needs to produce structured fields without losing the meaning of what the caregiver said. Documentation also supports continuity of care: when information transfer between care providers is incomplete or delayed, patient safety and continuity are affected [5].

2.2.2 Transformer models and transfer learning

Most modern speech and language models are based on the Transformer architecture, which uses self-attention instead of recurrence to model relationships within a sequence [11]. Two sequence-to-sequence Transformer models are used in this research. T5 frames every task as text-to-text: the input is a text with a task prefix and the output is text, which makes it possible to train one model to produce structured strings [12]. BART combines a bidirectional encoder with an autoregressive decoder and is pre-trained as a denoising autoencoder, by corrupting text and learning to reconstruct it [13]. Because removing disfluencies is a form of reconstructing clean text from noisy text, BART's pre-training objective is closely related to the auto-editing task in SCRIBE.

2.2.3 Speech recognition and parameter-efficient adaptation

Whisper is an encoder-decoder Transformer for speech recognition trained on 680,000 hours of weakly supervised multilingual audio [14]. It processes audio as a log-Mel spectrogram over a fixed 30-second window, so longer recordings must be split into segments. Its small variant has about 242 million parameters, which makes it practical for limited hardware. Fine-tuning all parameters of such a model is expensive and needs large amounts of data. Low-Rank Adaptation (LoRA) instead freezes the pre-trained weights and learns small low-rank update matrices that are added to selected layers [15]. The rank of these matrices controls how much the model can adapt and how many parameters are trained. LoRA therefore suits a setting such as this one, where the amount of accented training speech is limited and the deployment target is modest hardware.

2.2.4 Disfluencies and self-corrections

Spontaneous speech contains disfluencies such as filled pauses ("uh", "um"), repetitions and repairs. Shriberg described the structure of a repair as a reparandum (the part that is replaced), an interruption point, an optional editing term (for example "no wait" or "sorry") and the repair itself [16]. In "I gave her two, no wait, one tablet", "two" is the reparandum, "no wait" is the editing term and "one" is the repair. For documentation, the important point is that the repair carries the intended meaning. A system that only deletes filler words would keep both "two" and "one", and the record

could show the wrong dose. This is why self-correction handling is treated as a safety issue in SCRIBE.

2.3 Related Research

2.3.1 Speech recognition for accented English

Javed et al. introduced Svarah, a benchmark of 9.6 hours of English speech from 117 speakers across 65 locations in India, covering both read and spontaneous speech [6]. Their evaluation showed that existing English speech recognition systems leave clear room for improvement on Indian accents. This motivates accent adaptation for South Asian caregivers. Svarah was used in this research as the adaptation corpus because it is public, accent-specific and includes speaker diversity, although it is not caregiving speech.

Speech recognition errors matter more in health records than in everyday dictation. Zhou et al. analysed 217 dictated clinical notes and found an error rate of 7.4% in the drafts produced by speech recognition software, and 5.7% of these errors were judged clinically significant; most errors were corrected only after review by transcriptionists and physicians [17]. Koenecke et al. showed a different risk specific to Whisper: in about 1% of the transcriptions they examined, the model produced entire phrases that were not present in the audio, and a large share of these hallucinations contained harmful content [18]. Both findings support two design choices in SCRIBE: the raw transcript is not used as the final record but is cleaned and mapped to fixed fields, and the reports are produced from templates rather than generated as free text.

2.3.2 Disfluency correction

Bhat et al. released DISCO, a human-annotated disfluency correction corpus for English, Hindi, German and French, and reported that removing disfluencies improved downstream machine translation [9]. The English part contains disfluent and fluent sentence pairs from voice-assistant style commands. Wang and Herremans released DisfluencySpeech, a single-speaker re-recording of expressive utterances from the Switchboard corpus of spontaneous telephone conversations [19], with about 10 hours of speech and three transcript levels, ranging from a version that keeps all paralinguistic events to a clean version without non-sentence elements and false starts

[10]. Both corpora provide general-domain training pairs for disfluency removal, but neither contains caregiving vocabulary or medication self-corrections, which is why domain-specific pairs were added in this work.

2.3.3 Extracting ADL information from text

Wieland-Jorna et al. systematically reviewed NLP systems that extract information about ADL from EHRs and identified 22 studies describing 20 systems published between 2017 and 2022 [7]. The reported challenges were restricted access to EHR data, inadequate documentation, a lack of granularity in how activities were recorded, and small datasets. These findings support the idea that ADL extraction is feasible, but they also show two gaps that SCRIBE addresses: the input in all reviewed studies was written text, and the lack of accessible data limited the work.

Large language models (LLMs) have also been applied to clinical information extraction. Agrawal et al. showed that a general LLM, without clinical training, performed well at zero-shot and few-shot extraction from clinical text, including tasks that require structured output such as span identification and relation extraction [20]. Using such models directly with patient data, however, raises privacy concerns. Tang et al. addressed this by using an LLM to generate labelled synthetic clinical text and then fine-tuning a smaller local model on it, which improved named entity recognition and relation extraction over using the LLM directly and reduced the need to share patient data [21]. SCRIBE follows a similar approach: an LLM was used only to generate the ADL training dataset from de-identified information, and a small local model (T5-small) performs the extraction.

2.3.4 AI-assisted clinical documentation

Kanaparthy et al. reviewed real-world evidence on digital scribes that use ambient listening and generative AI to draft clinical notes [8]. Digital scribes reduced self-reported documentation time and improved clinician engagement, but standard burnout measures did not change and a few studies reported generated content that was not supported by the conversation, meaning that generated notes need review before they are finalised. This evidence is relevant in two ways. It shows the value of speech-based documentation, and it shows the risk of free-text generation in health records.

SCRIBE therefore extracts fields and produces reports from fixed templates rather than generating free text.

2.3.5 Grammar correction

LanguageTool is an open-source, rule-based style and grammar checker [22]. Rule-based correction is predictable and does not need training data, but it is designed for general text. Domain words such as medication names and local food names may be flagged as spelling errors and replaced, which would corrupt a care record. This problem was observed during testing and is addressed in SCRIBE with a protected vocabulary (Section 4.4.5).

2.4 Existing Systems, Products and Techniques

Table 2.1 compares the main approaches that a care home could use for daily documentation with the needs identified in the field study.

Table 2.1: Comparison of existing documentation approaches with SCRIBE requirements

Approach	Strengths	Limitations for care-home ADL documentation
Handwritten paper records (current practice at the care home)	No technology needed; familiar to staff	Time-consuming; inconsistent format; observations missed; hard to search or hand over
EHR systems with typed entry	Structured, searchable records	Typing takes time away from care; documentation burden remains [4]
General speech-to-text dictation	Fast input by voice	Produces a raw transcript; keeps fillers and self-corrections; weaker on South Asian accents [6]; no structured fields
Ambient AI scribes for clinicians	Reduce documentation time [8]	Designed for doctor consultations; free-text notes that need review; no ADL fields or caregiver alerts
NLP extraction from written EHR notes	Can extract ADL information [7]	Requires written notes; limited by data access and small datasets
Rule-based filler removal	Simple and transparent	Removes "uh" and "um" but cannot resolve self-corrections
SCRIBE (this work)	Voice input; accent adaptation; self-correction handling; 19-field ADL extraction; alert flag; text and audio handover	Trained mainly on public and generated data; needs validation with real caregiver speech

2.5 Gap Synthesis and Conceptual Framework

The reviewed work leaves four connected gaps. First, accented speech recognition has public benchmarks, but lightweight adaptation for caregiver use has not been studied together with documentation needs. Second, disfluency correction corpora are general-domain and do not test medication doses or times, where a wrong repair is a safety problem. Third, ADL extraction research works on written EHR notes and is limited by data access [7]. Fourth, AI documentation tools focus on narrative notes for clinicians and do not decide which observation needs a supervisor's attention [8]. Table 2.2 summarises how SCRIBE addresses each gap.

Table 2.2: Research gaps and how SCRIBE addresses them

Gap in existing work	How SCRIBE addresses it	Evaluated in
Speech recognition is weaker on South Asian accents [6]	LoRA adaptation of Whisper-small on Indian-accented English with data-size and rank ablations	Section 6.1.1
Disfluency corpora lack caregiving self-corrections	General corpora combined with domain pairs; model choice based on safety-relevant corrections	Section 6.1.2
ADL extraction only from written EHR notes; little accessible data [7]	ADL extraction from spoken observations; de-identified dataset built from care-home records	Section 6.1.3
Documentation tools do not identify urgent observations	Alert flag in every record, a 200-case safety evaluation and field-based alert rules	Sections 6.1.4 and 6.1.5
Long, multi-observation recordings	30-second audio chunking and sentence segmentation	Section 6.2

The conceptual framework of SCRIBE follows directly from these gaps. Speech is first converted to text by an accent-adapted recogniser, then cleaned so that only the intended meaning remains, then mapped to a structured record, and finally checked for safety before it reaches the supervisor and the handover summary. Each stage reduces a specific kind of error: recognition errors, disfluency errors, structuring errors and missed alerts. Because each stage is a separate model or rule, it can be evaluated and improved on its own. The concrete architecture is presented in Section 4.2.

3 RESEARCH METHODOLOGY

3.1 Research Design

This research followed an applied, experimental design in which a working system was built and each part of it was evaluated against measurable criteria. The work combined qualitative field study, to understand how caregivers document care, with quantitative experiments, to train and compare models. The overall design is shown in Figure 3.1. It had five stages: a field study at a care home, data preparation, model development, evaluation, and integration into the CareSense platform. Data preparation, model development and evaluation were carried out in three parallel streams, one for each model in the pipeline. The speech recognition stream used the Svarah corpus to adapt Whisper-small with LoRA. The disfluency removal stream combined DisfluencySpeech, DISCO (English) and domain-specific caregiving pairs to compare T5-small and BART-base, and BART-base was selected. The ADL extraction stream used the de-identified ADL dataset built from the field study to compare T5-small and BART-base again, and T5-small was selected. The selected models were then combined into the SCRIBE pipeline and integrated with the CareSense platform.

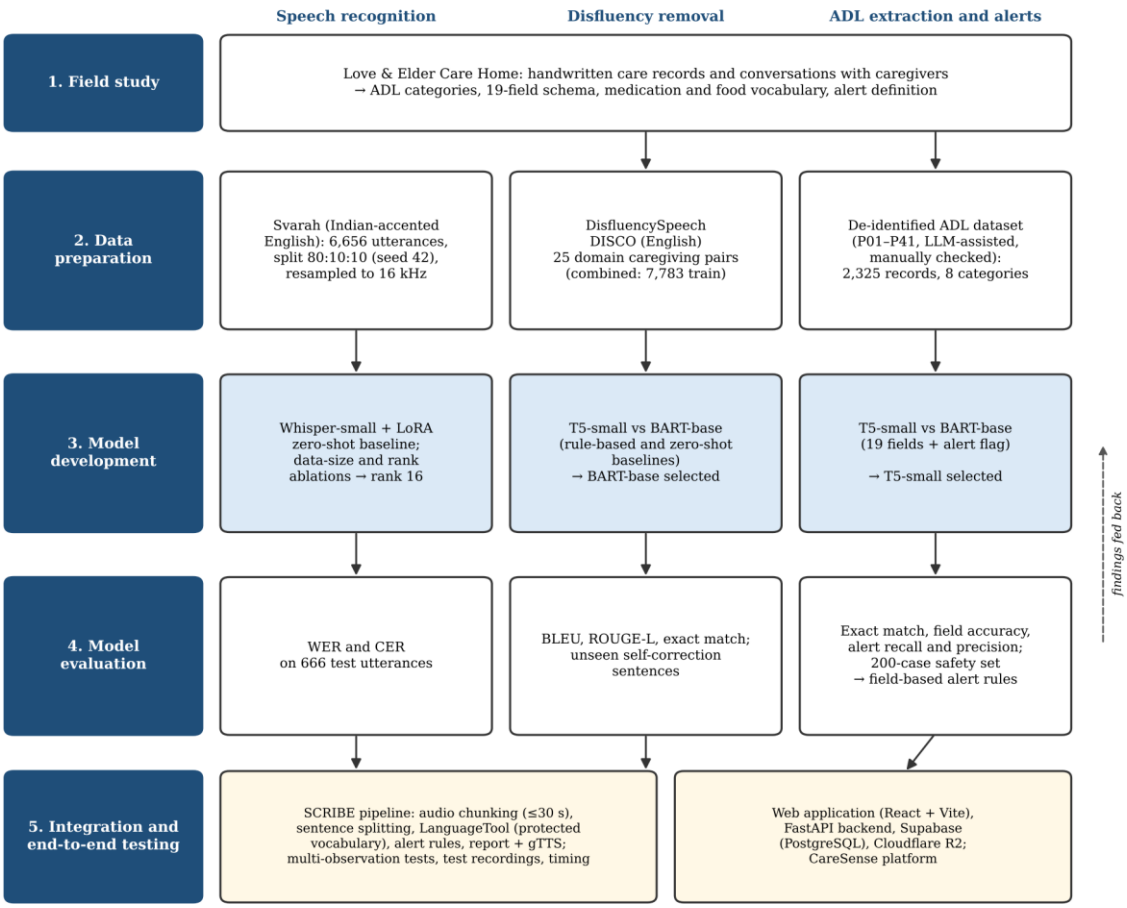


Figure 3.1: Research design of the SCRIBE component

The process was iterative. Evaluation results were used to change the development work, and several important improvements came from this loop. Qualitative tests revealed a data-ordering problem in the speech experiments, an output-parsing error in the extraction evaluation and a grammar tool that changed medication names. The panel comment at the first progress presentation (PP1), "improve and further validate the results", led to the 200-case safety evaluation, and the question about long recordings at the second progress presentation (PP2) led to the long-recording work. For each model, a zero-shot or rule-based baseline was measured first, so that every fine-tuned model could be compared with a clear starting point. Where two architectures were compared, they were trained on the same data and evaluated on the same test sets.

3.2 Data, Participants and Study Materials

3.2.1 Field study at the care home

The field study was carried out at the Love & Elder Care Home, a care home for older adults. Several visits were made. During these visits, the author collected the home's handwritten care records with the permission of the care home's management and talked with the caregivers, who explained how they record daily care and which observations they consider important. A second care home was also visited, and its staff described the same difficulties with handwritten records. No audio was recorded during the visits, and no care recipient was interviewed.

Each care recipient has an individual paper file, and the files are kept together in the office, as shown in Figure 3.2. The handwritten records, samples of which are shown in Figure 3.3, contain dated clinical notes such as hypertension and diabetes status, blood pressure readings (for example 160/98 mmHg and 171/110 mmHg), a random blood sugar reading, requested laboratory tests, and prescriptions with the medication name, strength and frequency (for example Aspirin 75 mg at night, Atova 10 mg at night and Losartan potassium 50 mg twice a day). The home also uses a weekly pill organiser and a printed medication chart. These records showed the vocabulary, abbreviations and medication names that caregivers use, and the kinds of observations that matter in practice.



Figure 3.2: Individual care recipient record files at the care home (care recipient name obscured)

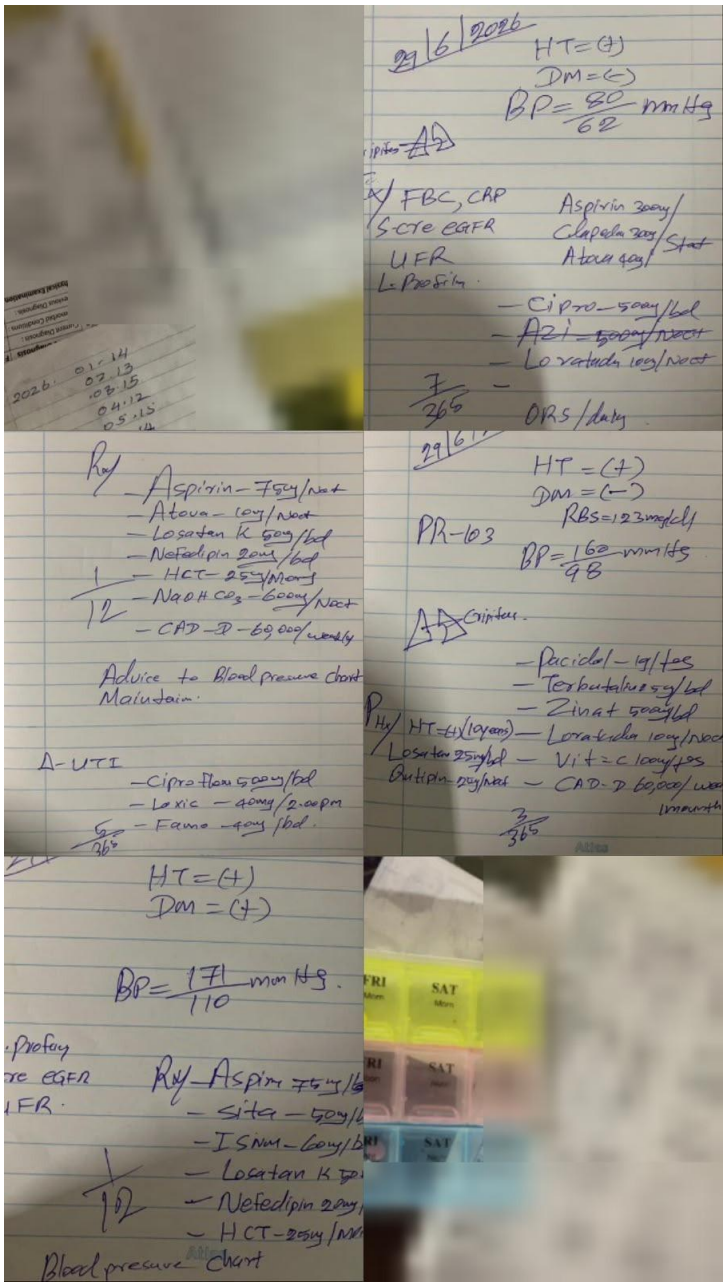


Figure 3.3: Samples of handwritten care records (care recipient names obscured)

The caregivers' explanations were used in three ways: to decide which ADL categories and fields the schema should contain, to define when an observation requires an alert, and to check the generated dataset against what caregivers actually record (Section 3.2.4).

3.2.2 Speech data

The Svarah corpus [6] was used to adapt and evaluate speech recognition. It contains 6,656 utterances (about 9.6 hours) of Indian-accented English from 117 speakers. The

corpus is distributed as a single test split, so it was divided at random into training, validation and test sets in an 80:10:10 ratio with a fixed seed of 42, giving 5,324, 666 and 666 utterances. All audio was resampled to 16 kHz, the input rate expected by Whisper. A check of 50 training utterances found a mean duration of 4.67 seconds, with a minimum of 0.37 seconds and a maximum of 13.10 seconds, so every utterance fitted within Whisper's 30-second window. Because the split was made by utterance and not by speaker, the same speakers can appear in both the training and test sets; this is discussed as a limitation in Section 7.5.

3.2.3 Disfluency data

Three sources were combined to train the disfluency-removal models, as summarised in Table 3.1. From DisfluencySpeech [10], the transcript that keeps fillers and false starts was used as the input and the fully cleaned transcript as the target. From DISCO [9], the 3,979 English disfluent-fluent sentence pairs were used and split 80:10:10 with seed 42. In addition, 25 domain-specific pairs were prepared for caregiving speech, such as "I gave her uh Metformin, 500... no wait, it's..." with its corrected form. Five of these pairs were held out as a domain test set, and the remaining 20 were repeated five times in the training set so that they were not lost among the much larger general-purpose data.

Table 3.1: Disfluency training and evaluation data

Source	Train	Validation	Test
DisfluencySpeech	4,500	250	250 (general test set)
DISCO (English)	3,183	398	398 (not used for the reported results)
Domain-specific caregiving pairs	20 × 5 = 100	–	5 (domain test set)
Total	**7,783**	**648**	–

3.2.4 ADL dataset

No public dataset exists for ADL extraction from caregiver speech, so a dataset was created for this research. The process is shown in Figure 3.4. First, the author manually typed the information from the handwritten records and the caregivers' explanations into a structured format grouped by medication regimens, meals, routines and symptoms. Care recipients' names and ages were left out at this stage, and each care

recipient was given a code from P01 to P41. The distribution of codes followed the home's real gender and medication pattern: 22 female care recipients (11 on medication and 11 not) and 19 male care recipients (all on medication).

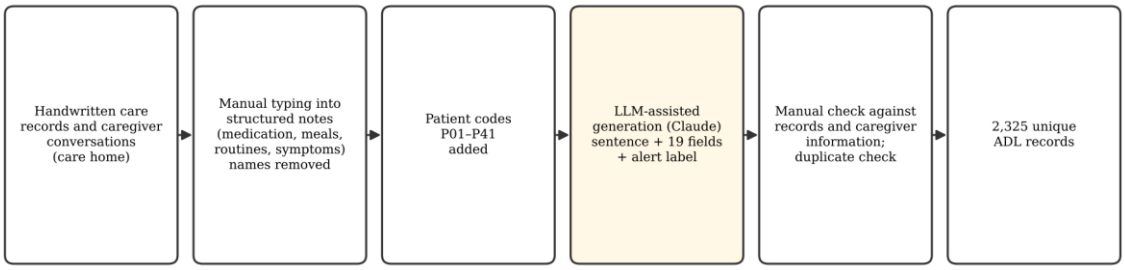


Figure 3.4: Creation of the de-identified ADL dataset

The typed, de-identified information was then given to Claude, a large language model developed by Anthropic [23], which was asked to generate unique caregiver-style observation sentences together with their 19 structured fields and an alert label, based only on the information provided. The alert rules given to the model were the ones the caregivers described, for example that any reported symptom, a refused or partially eaten meal, or an elevated vital-sign reading needs the supervisor's attention. The author then checked all generated records against the handwritten records and the caregivers' explanations, and confirmed that there were no exact duplicate sentences. The final dataset has 2,325 unique records in eight categories (Table 3.2). The full 19-field schema is given in Section 4.4.6 and in Appendix A.

Table 3.2: Categories in the ADL dataset

Category	Records	Example of category-specific fields
meal	500	food, meal type, intake level
medication	400	medication name, dosage
hygiene	300	hygiene activity
mobility	280	mobility type, destination
symptom	280	symptom type
fluid_intake	220	fluid type, amount
nurse_check	200	vital type, reading, status
family_visit	145	visitor type, visit reason
Total	**2,325**	

For training, each record was converted into one target string that lists all 19 fields in a fixed order, for example "category: symptom | medication: | dosage: | ... | symptom: headache | ... | time: evening | alert: Yes" for the sentence "P23 complained of headache after lunch." Fields that do not apply to a category are left empty. The dataset was split 80:10:10 with seed 42 into 1,860 training, 232 validation and 233 test records. The test split contains 54 records that require an alert.

3.2.5 Safety-focused test set

Because the original test split comes from the same generation process as the training data, a separate safety-focused test set of 200 sentences was constructed after the PP1 panel asked for further validation. It was never used for training. It contains 110 symptom sentences (five different phrasings for each of the 22 symptom types used in the dataset), 20 meal sentences describing partial or refused intake, 20 nurse-check sentences with elevated blood pressure or blood sugar, and 50 non-alert control sentences covering normal meals, normal vital signs, routine medication, mobility and hygiene (Table 3.3). The expected alert label for every sentence was fixed before the model was run, and the labels were reviewed by a caregiver.

Table 3.3: Composition of the 200-case safety-focused test set

Group	Cases	Expected alert	Content
Symptom alerts	110	Yes	22 symptom types × 5 phrasings
Meal alerts	20	Yes	10 partial intake, 10 refused meals
Vital-sign alerts	20	Yes	10 elevated blood pressure, 10 elevated blood sugar
Non-alert controls	50	No	10 each: normal meals, normal vitals, routine medication, mobility, hygiene
Total	**200**	150 Yes / 50 No	

3.2.6 Test recordings

To test the complete pipeline with realistic input, the author recorded caregiver observations by naturally speaking them in simple English, in the way a caregiver would describe a care recipient's daily care. The recordings cover everyday ADL observations such as medication, bathing, meals, drinking water, moving to the dining room, nurse checks and family visits. The recordings used after PP2 are referred to in this thesis as Run 1, which is 30.8 seconds long and contains six observations, and Run

2, which is 51.0 seconds long and contains nine observations. In Run 2, the blood-pressure and chest-pain observations were spoken after the 30-second point, so the recording could also be used to test the effect of Whisper's 30-second window. A longer recording describing three activities (morning hygiene, breakfast and medication) was used to examine how the pipeline from PP2 handled multi-observation input (Section 6.2.1).

3.3 Experimental Procedure

3.3.1 Speech recognition

A zero-shot baseline was first measured by running the pre-trained Whisper-small model [14] on the 666 test utterances. LoRA adapters [15] were then trained on the query and value projection layers of the attention blocks using the PEFT library [24], with the LoRA scaling factor set to four times the rank and a dropout of 0.05. Two experiments were run, each changing one factor. Experiment A fixed the rank at 8 and trained on 10%, 25%, 50% and 100% of the training set. Experiment B used the full training set and compared ranks 4, 8, 16 and 32. Every model was trained for three epochs with a batch size of 8 and mixed precision, and evaluated on the same test set.

The first run of Experiment A produced an unexpected spike at 50% of the data (32.20% WER), and re-running the same setting gave 22.43%. Investigation showed that the subsets had been taken from the start of the training set in its original order, so they were not representative. The training set was therefore shuffled with seed 42 before taking subsets, and Experiment A was repeated; only the shuffled results are reported. The final model was retrained with the selected rank as a separate run.

3.3.2 Disfluency removal

Three baselines were measured on the general and domain test sets: a rule-based method that deletes the filler words "uh", "um", "ah", "hmm", "er" and "erm", and zero-shot T5-small and BART-base. T5-small [12] and BART-base [13] were then fine-tuned on the same combined training data for three epochs with a batch size of 8. T5-small used the input prefix "clean:" and a learning rate of 3×10^{-4}; BART-base used a learning rate of 2×10^{-5} with a weight decay of 0.01, the commonly used settings for each model. Both models were evaluated on the same test sets and then on

about 28 hand-written, unseen caregiver sentences containing self-corrections of doses, times, foods and activities.

3.3.3 ADL extraction

T5-small was fine-tuned with the input prefix "extract:" for five epochs with a learning rate of 3×10^{-4}, a batch size of 8 and a maximum length of 192 tokens, and evaluated with greedy decoding on the 233-record test split. BART-base was trained with a learning rate of 2×10^{-5} for five epochs and evaluated with four-beam search, and was then retrained for 12 epochs to give it a further chance to learn the long target format. Both models were also tested on hand-written sentences that use different phrasing from the dataset. During evaluation, an error in the function that parses the generated field string was found: it erased the category whenever the category value was also a field name (for example "hygiene" or "symptom"), affecting 154 of the 233 test records. The parser was corrected before the reported T5 results were produced.

3.3.4 Safety-focused evaluation

The final T5-small model was run, without retraining, on the 200-case safety set. For each case the predicted alert was compared with the expected label, and all false negatives and false positives were inspected with their extracted fields. A separate out-of-scope probe described dizziness, which is not one of the 22 symptom types in the dataset.

3.3.5 Field-based alert rules

The error analysis showed that most missed alerts were records in which T5-small extracted the category and fields correctly but set the alert flag wrongly. A rule layer was therefore added after T5-small that sets the alert from the extracted fields according to the alert definition given by the caregivers (Section 4.4.7). The rules were evaluated by re-scoring the stored T5-small outputs for the 200 safety cases, so no model was retrained. A simpler variant, in which any record in the symptom category raises an alert, was also tested as an additional analysis.

3.3.6 Long recordings and processing time

Two changes were made after PP2: the audio is cut into equal pieces of at most 30 seconds before transcription, and the transcript is split into sentences before disfluency

removal and extraction. Their effect was measured in three ways. First, 100 multi-observation paragraphs were built by joining two or three randomly chosen sentences from the ADL test split (seed 42), giving 243 events of which 64 need an alert, and extraction was run on each whole paragraph and on its split sentences. Second, the old and new pipelines were run on the Run 1 and Run 2 recordings. Third, the complete new pipeline, including the alert rules, was timed on the Run 2 recording on a Google Colab T4 GPU (three runs) and on a Colab CPU runtime (two runs), excluding model loading and text-to-speech.

Table 3.4: Main training settings

Model	Task	Key settings
Whisper-small + LoRA	Speech recognition	LoRA on q_proj and v_proj; rank 4/8/16/32 (final 16); alpha = 4 × rank; dropout 0.05; 3 epochs; batch 8; fp16
T5-small	Disfluency removal	Prefix "clean:"; learning rate 3e-4; 3 epochs; batch 8; max length 128
BART-base	Disfluency removal (final)	Learning rate 2e-5; weight decay 0.01; 3 epochs; batch 8; max length 128; 4-beam search in the pipeline
T5-small	ADL extraction (final)	Prefix "extract:"; learning rate 3e-4; 5 epochs; batch 8; max length 192; greedy decoding
BART-base	ADL extraction (compared)	Learning rate 2e-5; weight decay 0.01; 5 and 12 epochs; batch 8; max length 192; 4-beam search

3.4 Measures and Evaluation Metrics

Each stage was evaluated with metrics that fit its task.

- **Word error rate (WER) and character error rate (CER)** for speech recognition. WER = (S + D + I) / N, where S, D and I are the numbers of substituted, deleted and inserted words and N is the number of words in the reference; CER is the same measure over characters. Both were computed with the JiWER library [25] on the raw transcripts.
- **BLEU** [26] and **ROUGE-L** [27] for disfluency removal, which measure n-gram and longest-common-subsequence overlap between the cleaned output and the reference.

- **Exact match (EM)**, the share of outputs that are identical to the reference after lower-casing and whitespace normalisation. It is stricter than BLEU and matters for short care statements, where one wrong word can change a dose or time.
- **Field-level accuracy** for ADL extraction, the share of records in which each of the 19 fields matches the reference. This shows which fields are unreliable, which an overall score can hide.
- **Alert recall and alert precision.** Recall = TP / (TP + FN) is the share of alert-worthy observations that were flagged; precision = TP / (TP + FP) is the share of raised alerts that were correct. The F1-score and overall accuracy were also reported. Alert recall was treated as the most important metric, because a missed alert in a care home is more dangerous than a false alarm, which a supervisor can simply dismiss.
- **Event recall** for long recordings, the share of observations in a multi-observation input that appear as a record with the correct category.
- **Processing time**, the end-to-end time to process a recording, compared with the length of the recording.

3.5 Data Analysis

The analysis combined quantitative comparison with error analysis. For each stage, the fine-tuned models were compared with the baselines and with each other on the same test sets. The results were then inspected case by case, especially where numbers looked suspicious or where a model's output had safety implications. Model selection followed a safety-weighted criterion: when two models had similar overall scores, the one that handled clinically significant cases correctly was preferred, and when one model was clearly better on every measure, it was chosen directly. Rank selection for speech recognition also considered the number of trainable parameters, because of the lightweight design goal.

Two limitations of the analysis should be noted here. First, rank and data size were compared on the test set rather than on the validation set, so the selected configuration may be slightly optimistic. Second, each configuration was trained once; re-running identical settings showed differences of about 0.1 to 0.2 percentage points in WER, so

very small differences between configurations should not be over-interpreted. Both points are discussed further in Section 7.5.

3.6 Ethics, Privacy, Security and Responsible AI

The care home is a setting with vulnerable older adults, so privacy was a central concern throughout the research. The handwritten records were collected with the permission of the care home's management. An ethics application for the research was submitted to SLIIT; at the time of writing, a decision had not yet been received. All work described in this thesis therefore used de-identified information only.

- **De-identification before processing.** Care recipients' names and ages were removed when the author typed the handwritten information, and each care recipient was given a code (P01 to P41). Only this typed, de-identified information was given to the language model used to generate the dataset; no photographs of records and no names were shared with any AI tool.
- **Photographs.** The photographs in Figures 3.2 and 3.3 were edited to obscure care recipient names before inclusion in this thesis.
- **No speech from care recipients.** No audio was recorded at the care home. The test recordings were spoken by the author.
- **Legal context.** Sri Lanka's Personal Data Protection Act, No. 9 of 2022, treats data concerning health as a special category of personal data [28]. A deployed system must therefore obtain consent, restrict access to records and store data securely (Section 8.4).
- **Responsible use of AI outputs.** Reports and handover summaries are produced from fixed templates using only the extracted fields, so the system does not generate unsupported clinical content. Alerts support the supervisor and do not replace staff judgement, and the evaluation deliberately focused on missed alerts, the most harmful type of error.

3.7 Reproducibility and Research Artefacts

All experiments were run in Google Colab notebooks [29], and the datasets, trained models and results were saved to a project folder in Google Drive. Fixed random seeds (42) were used for every data split, shuffle and sampling step, and every experiment

wrote its results to CSV files. GPU training is not fully deterministic, so repeated runs can differ slightly, as noted in Section 3.5. Table 3.5 lists the main research artefacts.

Table 3.5: Main research artefacts

Artefact	Content
Data preprocessing notebook	Svarah loading, audio decoding fix, 80:10:10 split
ASR LoRA experiments notebook	Zero-shot baseline, Experiments A and B, final rank-16 adapter
Disfluency data and training notebooks	Combined dataset, baselines, T5 and BART fine-tuning, unseen-sentence tests
ADL data preparation and training notebooks	19-field target strings, 80:10:10 split, T5 and BART training and evaluation
Safety-focused evaluation notebook	200-case safety set, metrics, false-negative and false-positive analysis
Report and TTS notebook	Template report, spoken summary, gTTS audio
End-to-end and long-recording notebooks	PP2 pipeline test, chunking and splitting, 100-paragraph evaluation, Run 1 and Run 2, alert rules
CPU timing notebook	Stage-by-stage CPU timing of the full pipeline
Trained models	final_whisper_lora_adapter, final_bart_disfluency, STAGE5_RICH_FINAL_MODEL
Result files	CSV files of all experiments, safety results and multi-observation details

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements were derived from the research objectives, the field study and the feedback received at the progress presentations. Table 4.1 lists the functional requirements and Table 4.2 the non-functional requirements.

Table 4.1: Functional requirements

ID	Requirement	Source
FR1	The caregiver can select a care recipient and record a voice observation through a web application.	Field study
FR2	The system transcribes the recording, including recordings longer than 30 seconds.	Objective 2; PP2 feedback
FR3	The system removes fillers and repetitions and keeps only the corrected meaning of spoken self-corrections.	Objective 3
FR4	The system corrects grammar without changing medication or food names.	Testing
FR5	The system extracts one structured record per observation using the 19-field ADL schema.	Objective 4
FR6	The system flags observations that need the supervisor's attention.	Field study (caregivers' alert rules)
FR7	Records and alerts are stored and shown on the supervisor's web dashboard.	CareSense integration
FR8	The system produces a written report and a spoken handover summary.	Objective 5

Table 4.2: Non-functional requirements

ID	Requirement	Rationale
NFR1	Privacy: care recipients are identified only by codes; health data is stored securely.	Vulnerable care recipients; PDPA [28]
NFR2	Lightweight models that can be hosted without large infrastructure.	Resource-constrained care homes
NFR3	Processing faster than the length of the recording on the target hardware.	Records should be ready soon after speaking
NFR4	Always available on a shared computer, without frequent phone use.	Caregivers should not be seen using phones during care
NFR5	No free-text generation in records or summaries.	Avoid unsupported content in health records [8]
NFR6	Modular stages that can be evaluated and replaced separately.	Maintainability; ISO/IEC 25010 [30]

4.2 Component Architecture Within the Complete Solution

Figure 4.1 shows the architecture of SCRIBE and how it connects to the shared CareSense data layer. The caregiver uses a web application built with React and Vite [31], [32]. The application sends the voice log, together with the selected care recipient code, to a backend built with FastAPI [33], which runs the processing pipeline. The audio clip is stored in Cloudflare R2 object storage [34], and the structured ADL record and any alert are stored in the shared Supabase PostgreSQL database [35], which also holds the caregiver and patient records used by CareSense. The supervisor's web dashboard reads the records and alerts from this database and shows the reports.

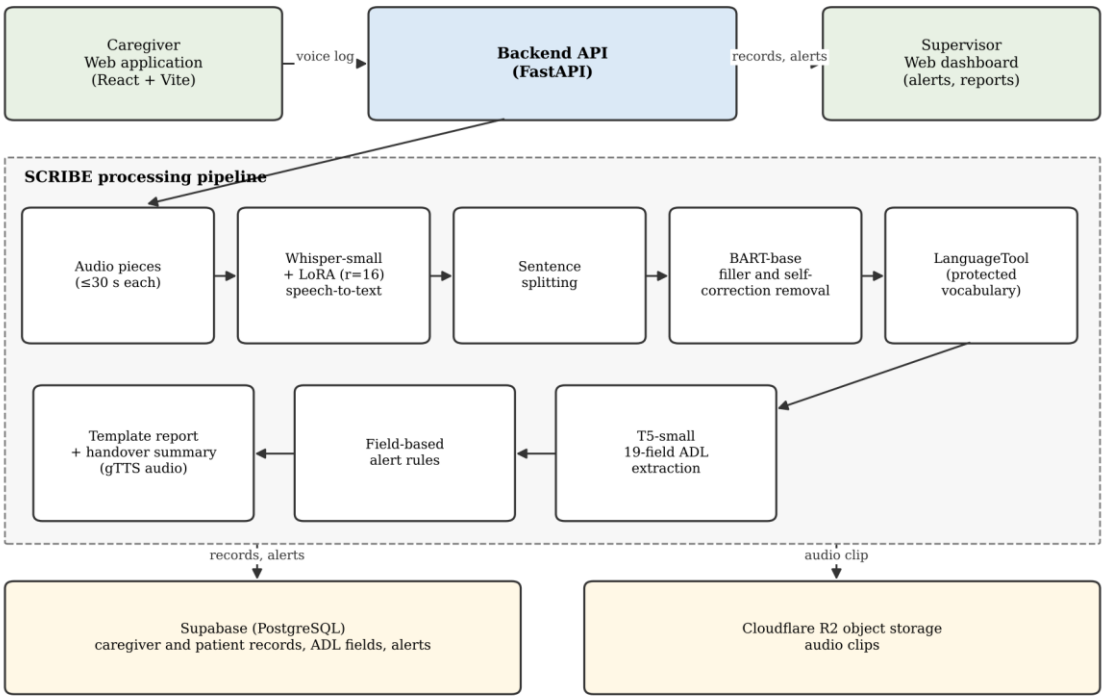


Figure 4.1: Architecture of SCRIBE and its connection to the CareSense data layer

The pipeline has eight stages. The audio is cut into pieces of at most 30 seconds and transcribed by Whisper-small with the LoRA adapter. The transcript is split into sentences, and each sentence is cleaned by BART-base and checked by LanguageTool with a protected vocabulary. T5-small then extracts the 19-field record, the alert rules decide the alert flag, and fixed templates produce the written report and the spoken handover. Keeping each stage separate means that one model can be replaced without retraining the rest of the pipeline.

4.3 Design Alternatives and Decisions

Several design alternatives were considered. The main decisions and the evidence behind them are summarised in Table 4.3.

Table 4.3: Main design alternatives and decisions

Decision	Alternatives considered	Choice and reason
Client platform	Mobile app; web app	Web app. Caregivers cannot use their phones often during care, so a web application that stays open on a shared computer is more practical.
Speech model	Larger Whisper models; full fine-tuning; LoRA on Whisper-small	Whisper-small with LoRA: lightweight and needs little accented training data (Section 6.1.1).
LoRA rank	4, 8, 16, 32	Rank 16: close to the best WER (rank 32) with half the trainable parameters.
Disfluency model	Rule-based; T5-small; BART-base	BART-base: similar overall scores to T5-small but correct on more safety-relevant self-corrections.
Extraction model	T5-small; BART-base	T5-small: clearly better on every measure in the experiments.
Long recordings	Retrain T5 on multi-observation inputs; split into sentences	Sentence splitting: matches how T5 was trained and keeps the evaluated model unchanged.
Alert decision	Model flag only; field-based rules	Rules on the extracted fields: corrected most alert-flag errors (Section 6.1.5).
Report and handover	Generative summary; fixed templates	Templates: predictable and cannot add unsupported content.
Speech output	Recorded voice; text-to-speech	gTTS with an Indian English voice: simple to integrate for a prototype.

4.4 Detailed Design

4.4.1 Audio chunking

Whisper processes a fixed 30-second window [14]. When a longer recording is passed to the feature extractor in one piece, everything after 30 seconds is silently discarded. SCRIBE therefore divides the recording into the smallest number of equal pieces that are each at most 30 seconds long. For example, a 51-second recording becomes two pieces of 25.5 seconds. Equal pieces avoid a very short final piece, which in testing caused the model to output a stray non-English character. Each piece is transcribed separately, characters outside the basic Latin range are removed, and the texts are joined.

4.4.2 Speech recognition

The speech recognition stage uses the Whisper-small model with the rank-16 LoRA adapter applied to the attention query and value projections. Only 1,769,472 parameters are trained, about 0.7% of the 241.7 million parameters of Whisper-small, so the adapter file is small and the base model can be shared.

4.4.3 Sentence segmentation

The transcript is split into sentences after a full stop, question mark or exclamation mark, but only when the next word starts with a capital letter. This rule keeps times such as "7.30 am" together. Fragments with fewer than two words are discarded, so very short noise fragments are dropped while short alert sentences such as "She vomited." are kept. Each sentence is then processed separately, because the extraction model was trained on one observation per input.

4.4.4 Disfluency removal

Each sentence is passed to the fine-tuned BART-base model, which outputs the cleaned sentence. For example, "I gave her two— uh, no wait, sorry, one tablet of Metformin." becomes "I gave her one tablet of Metformin." In the pipeline, four-beam search is used for decoding.

4.4.5 Safe grammar correction

LanguageTool [22] is run on each cleaned sentence. Any suggested correction that touches a protected word is ignored. The protected list contains the medication names (Metformin, Losartan, Atova, Aspirin, Sindopa, Glyco, Omez, Empa and Sita) and local food names (Kiribath, string hoppers, thosa, rotty, pittu, kadala and kawpi) taken from the care home's information. Without this filter, the grammar tool replaced some of these names with ordinary English words.

4.4.6 ADL schema and extraction

The extraction model receives the sentence with the prefix "extract:" and generates a string with all 19 fields in a fixed order, separated by vertical bars. A parser converts this string into a dictionary. Table 4.4 shows the schema. The schema was designed to capture category-specific detail, such as the specific food item or the specific vital-sign reading, rather than only whether an activity took place.

Table 4.4: The 19-field ADL schema

Field(s)	Used for	Example values
category	All records	meal, medication, hygiene, mobility, symptom, fluid_intake, nurse_check, family_visit
medication, dosage	medication	Losartan; 1 tablet
food, meal_type, intake	meal	Kiribath; breakfast; full / partial / refused
fluid, amount	fluid_intake	water; 250ml
hygiene	hygiene	sponge bath, diaper change, teeth brushing
mobility, destination	mobility	bed-to-chair transfer; dining area
symptom	symptom	chest pain, headache, vomiting
vital, reading, status	nurse_check	blood pressure; 158/95; elevated
visitor, reason	family_visit	sister; to check on her recovery
time	All records	morning, lunch, early evening, varies
alert	All records	Yes / No

4.4.7 Alert rules

After extraction, the alert flag is set from the extracted fields using the rules in Table 4.5, which follow the alert definition given by the caregivers. For categories not covered by a rule, the model's own flag is kept.

Table 4.5: Field-based alert rules

Condition on extracted fields	Alert
category = symptom and a symptom is named	Yes
category = meal and intake = partial or refused	Yes
category = meal and intake = full	No
category = nurse_check and status or reading contains elevated, high, low or abnormal	Yes
category = nurse_check and status = normal	No
Any other case	Model's predicted flag

4.4.8 Reports and spoken handover

Each record is turned into one report line using a template for its category, for example "Medication: Losartan given at morning." or "Symptom reported: chest pain at lunch. [ALERT]". Empty fields are shown as "unspecified" so that missing information is visible. The report ends with a summary of all alerts. The spoken handover uses short sentences such as "Patient P05 received Losartan this morning.", and states that an observation has been flagged for review only when it carries an alert. The handover text is converted to speech with gTTS [36] using the Indian English voice, so that the summary can be played at shift change.

4.5 Standards and Professional Practice

The ISO/IEC 25010 product quality model [30] was used as a checklist for the non-functional requirements: functional suitability (correct records and alerts), performance efficiency (processing time), maintainability (modular stages), security (protected health data) and usability (voice input and simple reports). Data handling was designed with Sri Lanka's Personal Data Protection Act in mind [28]. All models and libraries used are open source, and their licences must be checked before commercial use (Section 8.4). References follow the IEEE style recommended for the module, and all experiments are recorded in notebooks with fixed seeds so that they can be reviewed and repeated.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Model development and evaluation were carried out in Python in Google Colab notebooks [29] with an NVIDIA T4 GPU, and the datasets, models and results were stored in Google Drive. The application was built as a web application with a React and Vite front end, a FastAPI backend, a Supabase PostgreSQL database and Cloudflare R2 object storage. Table 5.1 lists the main technologies.

Table 5.1: Technology stack

Layer	Technology	Use in SCRIBE
Deep learning	PyTorch [37]; Hugging Face Transformers [38]	Model loading, fine-tuning and generation
Parameter-efficient tuning	PEFT [24]	LoRA adapters for Whisper-small
Data	Hugging Face Datasets [39]; pandas	Loading Svarah and DisfluencySpeech; splits; Excel and CSV data
Audio	librosa; ffmpeg; openai-whisper (baseline)	Resampling to 16 kHz; converting uploaded MP4 and other formats; zero-shot baseline
Evaluation	JiWER [25]; Hugging Face evaluate; scikit-learn	WER and CER; BLEU and ROUGE-L; confusion matrix
Grammar	LanguageTool via language-tool-python [22]	Rule-based grammar correction
Speech output	gTTS [36]	Spoken handover in Indian English voice
Front end	React with Vite [31], [32]	Caregiver web application and supervisor dashboard
Backend	FastAPI [33]	API that runs the processing pipeline
Storage	Supabase PostgreSQL [35]; Cloudflare R2 [34]	Structured records and alerts; audio clips
Compute	Google Colab T4 GPU and CPU runtimes	Training, evaluation and timing

5.2 Component Implementation

5.2.1 Speech recognition

The Svarah dataset was loaded from Hugging Face. By default, its audio column was decoded into a torchcodec decoder object rather than an array, which could not be used directly for feature extraction. This was fixed by casting the column to the standard audio feature type at 16 kHz, after which every example contained an audio array and its sampling rate. A custom data collator pads the log-Mel input features and the label

token sequences separately and replaces padding in the labels with -100 so that it is ignored in the loss. A single training function, parameterised by the training subset and the LoRA rank, was used for all ablation runs so that only one factor changed between experiments. For inference, the processor's feature extractor converts each audio piece to input features and the adapted model generates the transcript.

5.2.2 Disfluency removal and grammar correction

The three disfluency sources were mapped to a common pair of columns (raw text and clean text) and concatenated. Both T5-small and BART-base were trained with the Hugging Face Trainer, and the trained models were saved together with their tokenizers. In the pipeline, the cleaned sentence is passed to a grammar function that retrieves LanguageTool's suggested corrections, removes any suggestion whose matched text contains a protected medication or food name, and applies the remaining corrections.

5.2.3 ADL extraction and alert rules

Each record in the ADL dataset was converted into a target string that lists the 19 fields, and the model was trained to generate this string from the sentence. A parser splits the generated string at the field separators and maps each key to its value. Its first version replaced separator characters too broadly and erased the category whenever the category value was also a field name. A check found that 154 of 233 test records had a blank category for this reason, and the regular expression was corrected so that only genuine stray separator characters before a field name are replaced. The alert rules are implemented as a small function that receives the parsed fields and returns the record with the alert set according to Table 4.5.

5.2.4 Long recordings, reports and speech output

The long-recording support consists of two functions. The chunking function loads the audio at 16 kHz, computes the number of pieces needed so that none exceeds 30 seconds, transcribes each equal piece and joins the cleaned texts. The segmentation function splits the joined transcript with the regular-expression rule described in Section 4.4.3. The new pipeline then processes each sentence through disfluency removal, grammar correction, extraction and the alert rules, keeps every record with a recognised category, and builds a combined report with one numbered line per record

and a final alert summary. The spoken handover joins the per-record sentences and is saved as an MP3 file with gTTS.

5.3 Integration With the Other Components

SCRIBE is integrated with CareSense through the shared data layer. The caregiver and patient records used by the group are held in Supabase PostgreSQL, and SCRIBE writes each extracted ADL record, with its fields and alert flag, to this database. Audio clips, which are unstructured data, are stored in Cloudflare R2 object storage. The supervisor dashboard is a web interface that reads the stored records and alerts. Because the other components use the same database and care recipient records, the documentation produced by SCRIBE is available to the wider CareSense platform rather than being kept in a separate system.

5.4 Verification and Software Testing

Testing was carried out at three levels: data and code checks, model evaluation on held-out and unseen data, and end-to-end tests of the complete pipeline on real recordings. The model evaluations are reported in Chapter 6. Table 5.2 summarises the main test cases and their outcomes.

Table 5.2: Main verification test cases

ID	Test	Expected result	Outcome
TC1	Svarah audio decoding and format	Audio arrays at 16 kHz	Failed at first (decoder object); passed after casting the audio column; 50 checked samples at 16 kHz
TC2	Speech data split	5,324 + 666 + 666 = 6,656	Passed
TC3	Parsing of the 19-field output	Category and fields recovered correctly	Failed at first (154/233 blank categories); passed after correcting the parser
TC4	Target length within model limit	Targets shorter than 192 tokens	Passed (longest of 200 checked: 74 tokens for T5, 88 for BART)
TC5	Duplicate sentences	No exact duplicates	Passed for the ADL dataset and the 200-case safety set
TC6	Protected vocabulary	Medication and food names unchanged by grammar correction	Passed for correctly transcribed names (for example Losartan, Kiribath)
TC7	Recording longer than 30 s (Run 2, 51.0 s)	All speech transcribed	Old method: 42 words; new method: 71 words (29 words recovered)
TC8	Short final audio piece (Run 1, 30.8 s)	No spurious output	Fixed-length pieces produced a 0.8 s piece and a stray character; equal pieces and character filtering removed it
TC9	Three observations in one input	Three records; chest-pain alert raised	Old pipeline: one incorrect record; new pipeline: three correct records with the alert

ID	Test	Expected result	Outcome
TC10	Sentence splitting on 100 test paragraphs	Each paragraph split into its sentences	Passed (100% correct splits)
TC11	Report and spoken handover consistency	Both state the same alerts	Failed at first (spoken text said "flagged" without an alert); passed after the template fix
TC12	GPU and CPU give the same output	Same records and alerts	Passed (9 records and 2 alerts on both)

5.5 Deployment and Operation

SCRIBE is used through a web browser, so the caregiver application can stay open on a shared computer in the care home throughout the day. A typical use follows these steps: the caregiver selects the care recipient, records a short observation or a longer note with one observation per sentence, and submits it; the backend processes the recording and stores the records; the supervisor sees new records and any alerts on the dashboard; and the report and spoken handover can be produced at shift change.

The models are loaded once when the backend starts; loading all models took about 50 seconds on a CPU. Processing time depends on the hardware. On a Colab T4 GPU, a 51-second recording was processed in 12.2 seconds on average, while on a single-thread Colab CPU runtime it took 77.2 seconds (Section 6.2.3). A GPU is therefore recommended for near real-time use, while CPU-only operation is possible but slower than real time. Caregivers should pause briefly between observations, because the sentence splitter relies on the punctuation that Whisper adds at pauses.

6 RESULTS AND VALIDATION

This chapter reports the results of the experiments described in Chapter 3. Section 6.1 presents the research results for each model, Section 6.2 the results of the complete product, Sections 6.3 and 6.4 the stakeholder and integration validation, and Section 6.5 summarises the findings against the research questions.

6.1 Research Results

6.1.1 Speech recognition

Table 6.1 summarises the speech recognition experiments. Zero-shot Whisper-small reached a WER of 26.13% and a CER of 12.85% on the 666 test utterances. With the LoRA rank fixed at 8, WER fell steadily as more training data was used, from 19.51% with 10% of the data to 15.57% with all of it (Figure 6.1). The largest gain came from the first 10% of the data (532 utterances), which already reduced WER by about a quarter compared with the zero-shot model, and the improvement became smaller as more data was added.

Table 6.1: Speech recognition: baseline, ablation and final model results

Experiment	Configuration	Trainable parameters	WER (%)	CER (%)
Baseline	Whisper-small, zero-shot	–	26.13	12.85
Data efficiency (rank 8)	10% of training data (532)	884,736	19.51	6.20
	25% (1,331)	884,736	17.32	5.78
	50% (2,662)	884,736	16.31	5.49
	100% (5,324)	884,736	15.57	5.22
Rank ablation (100% data)	Rank 4	442,368	16.04	5.34
	Rank 8	884,736	15.70	5.19
	Rank 16 (selected)	1,769,472	**15.35**	**5.10**
	Rank 32	3,538,944	14.86	5.15
Final model	**Rank 16, retrained**	1,769,472	**15.57**	**5.32**

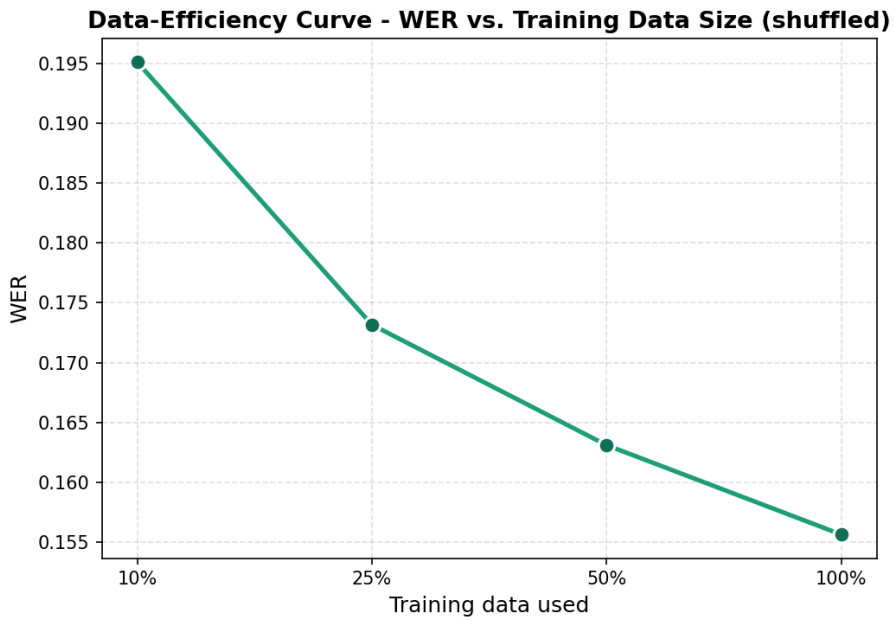


Figure 6.1: WER against the amount of training data (rank 8, shuffled subsets)

Across ranks, rank 32 gave the lowest WER (14.86%), and rank 16 came close (15.35%) with half the trainable parameters (Figure 6.2). Rank 16 was selected because of the lightweight design goal. When it was retrained as a separate run for the final model, it reached a WER of 15.57% and a CER of 5.32%, a 40.4% relative reduction in WER from the zero-shot baseline. The same configuration therefore gave 15.35% and 15.57% in two runs, and rank 8 with all the data gave 15.57% in Experiment A and 15.70% in Experiment B. These differences show that repeated runs vary by about 0.1 to 0.2 percentage points, so the gap between ranks 16 and 32 (0.49 points) is small. The adapter has only 1,769,472 trainable parameters, about 0.7% of the full model.

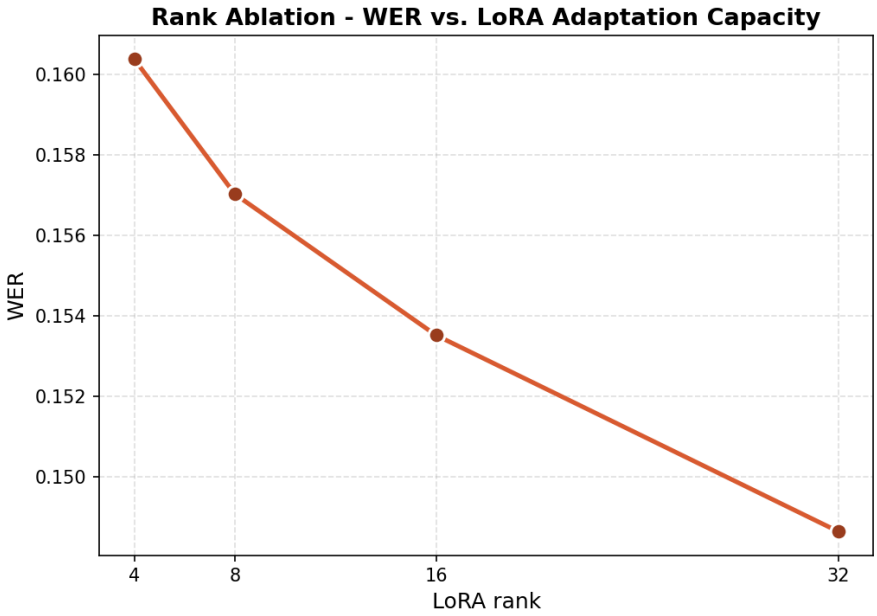


Figure 6.2: WER against LoRA rank (100% of training data)

6.1.2 Disfluency removal

Table 6.2 shows the disfluency removal results. On the general test set, the rule-based baseline already scored well on BLEU and ROUGE-L, because most words in a disfluent sentence are kept, but it produced an exact match for only 25.6% of sentences. Both zero-shot models were worse than the rule-based method. Fine-tuning improved both models substantially. T5-small scored slightly higher than BART-base on the general test set (exact match 67.2% against 64.0%, BLEU 0.9616 against 0.9530) and on the five-sentence domain test set (exact match 40% against 20%), while BART-base had a slightly higher domain ROUGE-L. Figure 6.3 compares the BLEU scores.

Table 6.2: Disfluency removal results

Model	General BLEU	General ROUGE-L	General EM (%)	Domain BLEU	Domain ROUGE-L	Domain EM (%)
Rule-based	0.8238	0.9444	25.6	0.3983	0.8164	0.0
T5-small (zero-shot)	0.5931	0.7804	1.6	0.2536	0.5850	0.0
BART-base (zero-shot)	0.7580	0.9252	14.8	0.3298	0.7524	0.0
T5-small (fine-tuned)	0.9616	0.9856	67.2	0.6111	0.8919	40.0
BART-base (fine-tuned)	0.9530	0.9817	64.0	0.5744	0.9053	20.0

General test set: 250 DisfluencySpeech sentences. Domain test set: 5 held-out caregiving sentences.

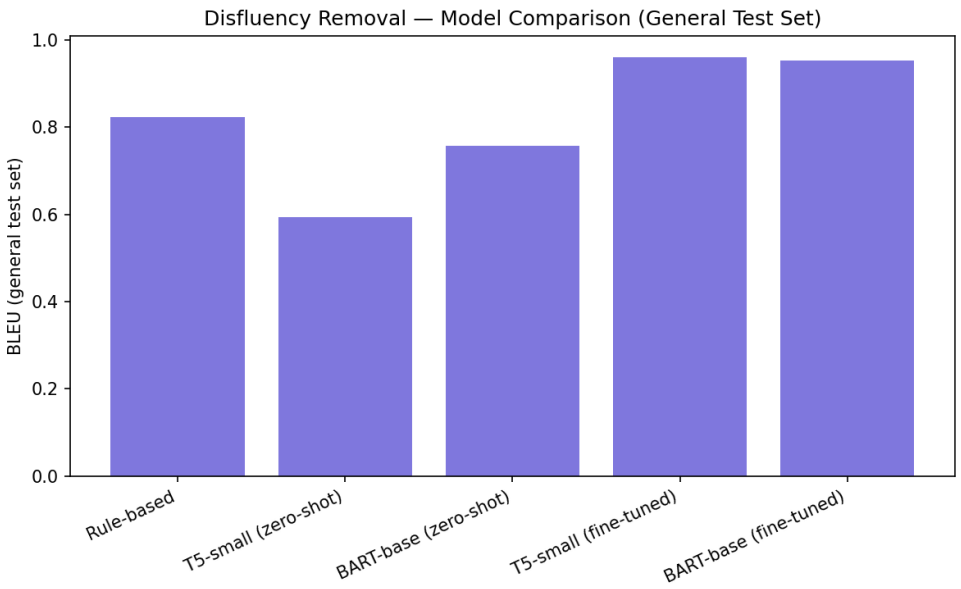


Figure 6.3: BLEU on the general test set for each disfluency removal approach

The aggregate scores did not separate the two fine-tuned models clearly, so the hand-written caregiver sentences were used to compare how they handle self-corrections. Table 6.3 shows representative cases. BART-base resolved five self-corrections that T5-small got wrong, including the medication dose case, in which T5-small kept the wrong dose ("two tablets") while BART-base kept the corrected dose ("one tablet"). T5-small resolved two cases that BART-base did not. Several food-related corrections, such as "milk tea, no wait, plain tea", were not resolved by either model. Because a wrong dose or time in a care record is a safety risk, BART-base was selected for the pipeline even though T5-small had slightly higher aggregate scores.

Table 6.3: Representative self-correction cases on unseen sentences

Input (caregiver-style speech)	T5-small output	BART-base output
I gave her two - uh, no wait, sorry, one tablet of Metformin.	I gave her two tablets of Metformin. ✗	I gave her one tablet of Metformin. ✓
Gave the medicine at, at 9... no actually it was 9:30am.	Gave the medicine at 9am. ✗	Gave the medicine at 9:30am. ✓
I gave the patient the, uh, wrong dose sorry, I mean the correct dose.	I gave the patient the, wrong dose sorry, the correct dose. ✗	I gave the patient the correct dose. ✓
...and I checked her at 2, no, actually 2:30pm.	...checked her at 2, no, actually 2:30pm. ✗	...checked her at 2:30pm. ✓
He needs help with bathing, uh, with dressing, actually.	He needs help with dressing, actually. ✓	He needs help with bathing, with dressing, actually. ✗
He ate rice for lunch, um, no, he only ate a few bites of rice.	He only ate a few bites of rice. ✓	He ate rice for lunch, no, he only ate a few bites of rice. ✗

Input (caregiver-style speech)	T5-small output	BART-base output
He drank plan tea, plan tea in the morning, no wait, it was milk tea at 9:30am.	Self-correction kept ✗	Self-correction kept ✗

6.1.3 ADL extraction

Table 6.4 compares the two extraction models on the 233-record test split. T5-small reached 100% accuracy on 16 of the 19 fields, 97.4% on intake, 99.6% on visit reason and 95.3% on time, and generated an exactly correct field string for 92.3% of records. It detected all 54 alert-positive records with no false alarms. BART-base performed much worse on the same task. After five epochs it detected only 20.4% of alerts and produced no exactly correct record. Retraining for 12 epochs raised alert recall to 70.4%, but free-value fields remained poor (Table 6.5), and its raw outputs contained malformed separators and unrelated words.

Table 6.4: ADL extraction results on the held-out test split (233 records, 54 alerts)

Model	Exact match (%)	Alert recall (%)	Alert precision (%)
T5-small, 5 epochs, greedy (selected)	**92.3**	**100.0 (54/54)**	**100.0**
BART-base, 5 epochs, 4-beam search	0.0	20.4 (11/54)	100.0
BART-base, 12 epochs, 4-beam search	0.0	70.4 (38/54)	100.0

Table 6.5: Selected field-level accuracy (%) on the test split

Field	T5-small	BART-base (12 epochs)
category	100.0	99.6
dosage	100.0	20.2
food	100.0	42.9
intake	97.4	39.9
amount	100.0	24.9
symptom	100.0	43.8
reading	100.0	27.9
status	100.0	24.9
time	95.3	71.7
alert	100.0	80.7

The same two architectures were therefore compared twice, and the deciding factor was different each time. For disfluency removal the choice was a narrow, safety-based trade-off, while for extraction it was a broad capability gap. However, BART-base was

trained with a lower learning rate than T5-small (2×10^{-5} against 3×10^{-4}), so part of this gap may come from the training settings rather than the architecture (Section 7.5).

Tests on hand-written sentences with different wording from the dataset showed that T5-small's near-perfect test scores do not fully transfer to new phrasing (Table 6.6). Typical errors were a dosage of "no" taken from "no issues", reversed or wrong intake levels, and treating dizziness as a mobility observation.

Table 6.6: Examples of T5-small errors on hand-written sentences

Sentence	Error in the extracted fields
Gave P05 her Losartan tablet this morning, no issues.	dosage = "no" (taken from "no issues")
P18 refused to eat lunch today and seemed upset.	intake = partial (should be refused)
P21 had bread with dhal curry for breakfast but left half of the meal unfinished.	intake = refused (should be partial)
P05 ate pasta with vegetables and chicken for dinner and appeared to have a good appetite.	intake = partial and alert = Yes (should be full, no alert)
P03 ate kiribath with lunu miris in the morning and seemed to enjoy it.	meal_type = "lunu miris" (a food, not a meal type)
P21 felt dizzy while walking to the dining area this afternoon.	Treated as mobility ("dizzy"); no alert

6.1.4 Safety-focused evaluation

On the 200-case safety set, T5-small reached an alert recall of 82.0% and a precision of 96.1% (Table 6.7, Figure 6.4). Of the 150 alert-worthy cases, 123 were detected and 27 were missed; of the 50 non-alert controls, 45 were correctly left without an alert and 5 were false alarms. Recall was 79.1% for symptoms, 85.0% for meal alerts and 95.0% for vital-sign alerts. The fall from 100% on the original test split to 82.0% on the safety set shows that a single test split from the same generation process overstates how reliably alerts are caught.

Table 6.7: Safety-focused alert evaluation of T5-small (200 cases)

Metric	Result
True positives / false negatives	123 / 27
False positives / true negatives	5 / 45
Alert recall / precision	82.00% / 96.09%
F1-score / accuracy	88.49% / 84.00%
Recall: symptom alerts	87/110 (79.1%)
Recall: meal-related alerts	17/20 (85.0%)
Recall: vital-sign alerts	19/20 (95.0%)
Specificity on non-alert controls	45/50 (90.0%)

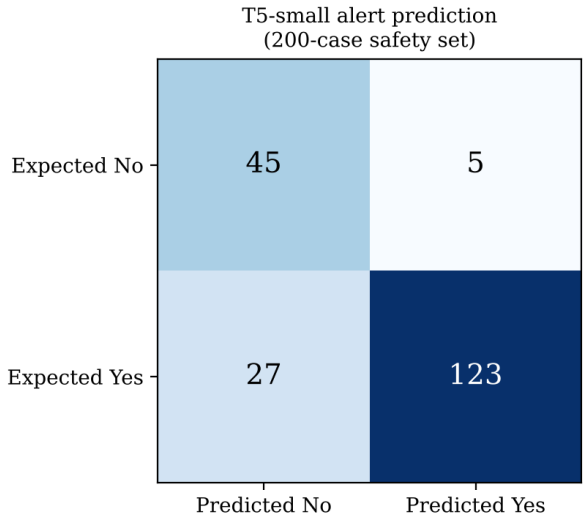


Figure 6.4: Confusion matrix of T5-small alert predictions on the safety set

Of the 27 missed alerts, 23 were symptoms, 3 were meals and 1 was a vital sign. Symptom recall varied by type: chest pain, headache, fever, breathing difficulty, body pain, inflammation and difficulty passing urine were detected in all five phrasings, while mild coughing and vomiting were detected in only two of five (the full table is in Appendix A). Inspecting the missed cases showed an important pattern: in the symptom misses, T5-small usually assigned the symptom category, and often the correct symptom name, but set the alert flag to "No". The five false alarms were four normal vital-sign readings and one normal meal; in the vital-sign cases the extracted status was "normal" but the flag was "Yes". In other words, most alert errors were cases where the alert flag contradicted the model's own extracted fields. The separate dizziness probe, an out-of-scope symptom, was extracted as a mobility observation and was not flagged.

6.1.5 Field-based alert rules

Because most alert errors came from the flag rather than from the extracted fields, the alert rules in Table 4.5 were applied to the stored T5-small outputs for the same 200 cases. The results are shown in Table 6.8 and Figure 6.5. The rules recovered 17 missed alerts (15 symptoms and 2 meals) and removed all five false alarms, raising recall from 82.0% to 93.3% and precision from 96.1% to 100%. The simpler variant, in which any symptom-category record raises an alert, raised recall to 98.7% at the cost of one false alarm.

Table 6.8: Alert detection on the safety set with and without alert rules

Method	Recall (%)	Precision (%)	FN	FP	Symptom recall (%)	Meal recall (%)	Vital recall (%)
T5-small only (model result)	82.0	96.1	27	5	79.1	85.0	95.0
T5-small + alert rules	93.3	100.0	10	0	92.7	95.0	95.0
T5-small + simpler symptom rule (additional analysis)	98.7	99.3	2	1	–	–	–

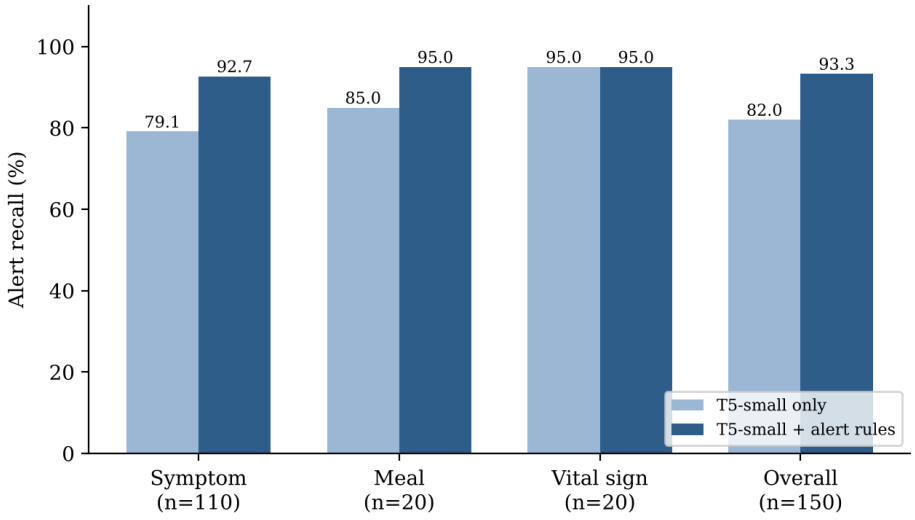


Figure 6.5: Alert recall by group for T5-small with and without the alert rules

The ten alerts still missed with the rules fall into three types. In eight cases (frequent urination, loose motion twice, skin allergy, speaking unrelated words twice and vomiting twice), T5-small assigned the symptom category but left the symptom name empty, so the rule, which requires a named symptom, did not fire. In one case the meaning was reversed: "P14 refused all of the food offered at breakfast" was extracted with intake "full". In one case, "a blood sugar reading of 185 mg/dL, flagged for follow-up", the sentence contained no alert keyword and the status was extracted as "normal". The single false alarm of the simpler variant came from a normal blood-pressure reading (115/65) that T5-small placed in the symptom category. The remaining errors are therefore extraction errors that rules cannot correct. Because the rules were designed after the errors were analysed and were tested on the same 200 cases, these improved figures are likely to be optimistic, and 82.0% remains the result of the trained model.

6.2 Product and Technical Results

6.2.1 Multi-observation inputs

At PP2, the system was demonstrated with a single observation that fitted within Whisper's 30-second audio window, and it processed this input correctly. The panel then asked how the system would handle longer recordings with several observations. When the PP2 pipeline was given a longer recording that described three activities (morning hygiene, breakfast and medication), it produced a single medication record with every field empty, and the spoken summary was "Patient P006 received unspecified this morning." Two causes were identified: the extraction model was trained on one observation per input but received the whole paragraph, and Whisper processed only the first 30 seconds of the audio.

Table 6.9 and Figure 6.6 show the effect of sentence splitting on 100 multi-observation paragraphs built from the held-out test split. Without splitting, event recall was 41.2%, which is the maximum possible when one record is produced for a paragraph that contains 243 events in 100 paragraphs; the model found one event in every paragraph and lost the rest. Record-level alert recall was only 18.8%. In some paragraphs the old method did raise an alert but attached it to the wrong record, for example flagging a medication record because the paragraph also mentioned a bed sore. With splitting, all 100 paragraphs were split correctly and event recall and alert recall both reached 100%. Because the paragraphs were built from test sentences that T5-small already handles well, this experiment shows that splitting recovers every observation, not that extraction is perfect on real speech.

Table 6.9: Extraction on 100 multi-observation paragraphs (243 events, 64 alert events)

Method	Event recall (%)	Alert recall (%)	Correct splits (%)
Whole paragraph (old)	41.2	18.8	–
Sentence splitting (new)	100.0	100.0	100.0

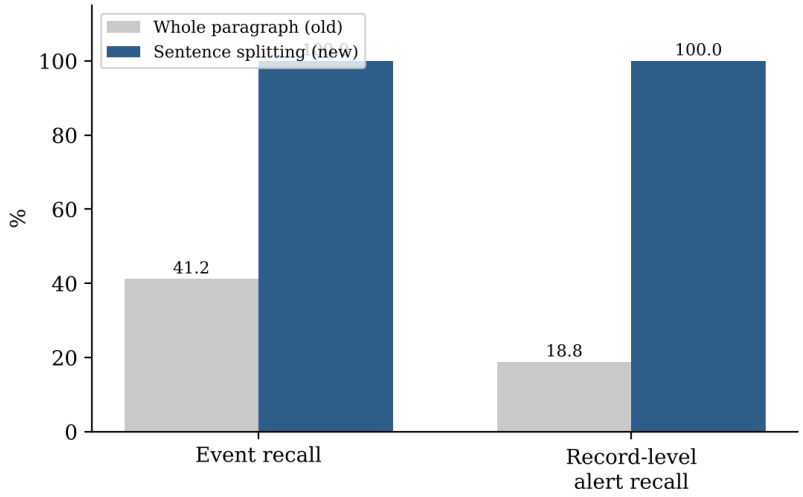


Figure 6.6: Event recall and alert recall on multi-observation paragraphs

6.2.2 Real recordings

On the Run 1 recording (30.8 seconds, six observations), the old pipeline produced one record that mixed fields from different observations (category meal, with "Kiribath" in the medication field and "1 glass" as the dosage). The new pipeline produced six records with five plausible categories. It also exposed speech recognition errors on words that matter: "Losartan" was transcribed as "Loosa Tant", which LanguageTool then changed to "Luisa Want" because the misrecognised name was not in the protected list; "nurse" became "nape"; and "high" became "I", so the blood-pressure observation was misclassified. The chest-pain observation was extracted correctly but was not flagged by the model.

The Run 2 recording (51.0 seconds, nine observations) was used to test all of the changes together (Table 6.10). The old method transcribed only the first 30 seconds (42 words) and lost the last three observations, including the blood-pressure and chest-pain alerts. The new method transcribed all 71 words, and the new pipeline produced nine records, eight with the correct category. Without the alert rules, neither alert was raised; with the rules, both the "very high" blood pressure and the chest pain were flagged, and the report and the spoken handover agreed.

Table 6.10: Results on the Run 2 recording (51.0 s, nine observations)

Measure	Old pipeline	New pipeline	New pipeline + alert rules
Words transcribed	42 (29 lost)	71	71

Measure	Old pipeline	New pipeline	New pipeline + alert rules
Records produced	1 (mixed fields)	9	9
Correct categories	–	8 of 9	8 of 9
Blood-pressure alert	Not heard	Missed	Raised
Chest-pain alert	Not heard	Missed	Raised

Table 6.11 lists the nine records produced with the alert rules. The remaining errors come from three sources: a speech recognition error ("brushed" heard as "pressed", so no hygiene activity was found), a cut in the middle of a sentence at the 25.5-second boundary ("She went to | in the dining room"), which turned the wheelchair observation into a meal record, and an invented dosage value ("april") for the medication record.

Table 6.11: Records extracted from the Run 2 recording with alert rules

#	Transcribed and cleaned sentence	Extracted record	Correct?
1	I gave Metformin tablet in the morning.	medication: Metformin; time: morning	Yes, but dosage "april" invented
2	She had bath at 8 o'clock.	hygiene: bath	Yes
3	She ate Kiribath at breakfast and finished it all.	meal: Kiribath, breakfast, full	Yes
4	She drank one glass of water.	fluid_intake: water, one glass	Yes
5	She pressed her teeth after the breakfast.	hygiene (activity empty)	Category only
6	She went to in the dining room in Wheelchair.	meal (fields empty)	No (should be mobility)
7	The nurse checked her blood pressure and it was very high.	nurse_check: blood pressure, very high; ALERT	Yes
8	After the lunch, she had chest pain.	symptom: chest pain; ALERT	Yes
9	Her daughter came to see her in the morning.	family_visit: daughter	Yes

6.2.3 Processing time

Table 6.12 reports the end-to-end processing time of the new pipeline, including the alert rules, for the 51.0-second Run 2 recording. On a Colab T4 GPU, the average over three runs was 12.2 seconds, about 4.2 times faster than real time. On a Colab CPU runtime, where PyTorch reported a single thread, the average over two runs was 77.2 seconds, about 1.5 times slower than real time. The CPU produced exactly the same records and alerts as the GPU. Figure 6.7 shows that speech recognition took 55% of the CPU time.

Table 6.12: End-to-end processing time for the 51.0-second recording

Hardware	Individual runs (s)	Average (s)	Compared with real time
Colab T4 GPU	15.1, 10.7, 10.9	12.2	4.2× faster
Colab CPU (1 thread)	86.9, 67.4	77.2	1.5× slower

Model loading (about 50 s on CPU) and text-to-speech are excluded. The first run on each device includes warm-up time.

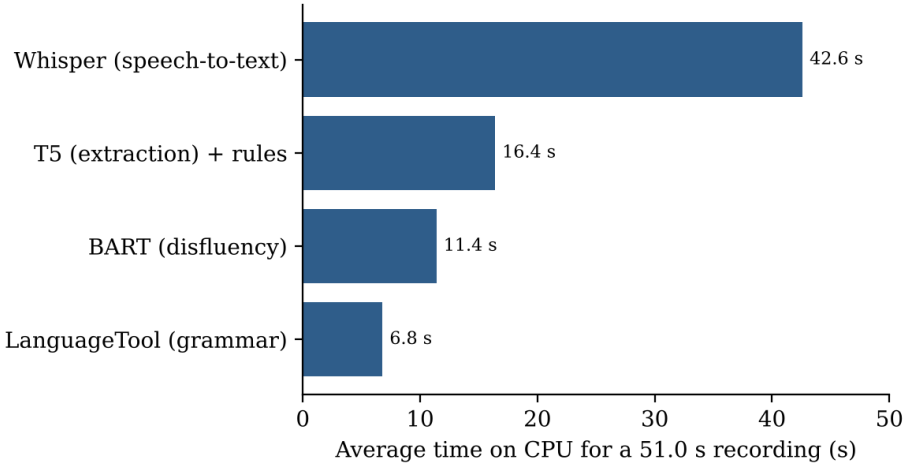


Figure 6.7: Average CPU time per pipeline stage for the 51.0-second recording

6.3 User or Stakeholder Validation

Stakeholder input shaped SCRIBE throughout the project, but a formal on-site evaluation with caregivers has not yet been completed. During the field visits, caregivers explained their documentation practice and the observations that require alerts, and these explanations defined the schema and the alert rules. When the system was explained to the staff of both care homes visited, they responded positively and expressed interest in using it. A caregiver reviewed the expected labels of the 200-case safety set. Caregivers who tried the system raised the question of how it would handle long, multi-sentence recordings, the same question raised at PP2, and this feedback led to the audio chunking and sentence splitting described in Section 6.2.

A structured evaluation of the system in the care home, with caregivers recording real observations and rating usability, is planned before the final viva and is identified as the most important next step. Until then, usability and real-world accuracy with spontaneous caregiver speech have not been measured quantitatively.

6.4 Integration Results

SCRIBE has been connected end-to-end with the CareSense platform through the shared data layer. Extracted ADL records with their alert flags are stored in the shared Supabase PostgreSQL database, which also contains the caregiver and patient records used by the group, and audio clips are stored in Cloudflare R2. The supervisor dashboard reads these records and alerts. Within SCRIBE, the full chain from audio to report and spoken handover was tested on the recordings described in Section 3.2.6, and the GPU and CPU runs produced identical outputs. Quantitative measures of the integrated platform, such as database write latency or dashboard update time, were not collected.

6.5 Summary of Findings

Table 6.13 summarises the main findings for each research question.

Table 6.13: Summary of findings by research question

Research question	Main finding
RQ1: accent adaptation of speech recognition	LoRA reduced WER from 26.13% to 15.57% (40.4% relative) with 0.7% of the parameters trained; most of the gain came from the first 10% of data; rank 16 was within about 0.5 points of rank 32.
RQ2: disfluency and self-correction handling	Fine-tuning raised exact match from 25.6% (rule-based) to 64–67%; BART-base was chosen because it resolved more safety-relevant self-corrections, although some food corrections remained unresolved.
RQ3: structured ADL extraction and alerts	T5-small reached 92.3% exact match and 100% alert recall on the test split but 82.0% recall on the 200-case safety set; field-based rules raised this to 93.3% with no false alarms.
RQ4: long recordings and processing time	Chunking and sentence splitting recovered all observations (event recall 41.2% to 100%); a 51-second recording took 12.2 s on GPU and 77.2 s on CPU.

7 DISCUSSION

7.1 Interpretation of Findings

Speech recognition. The results show that a lightweight model can be adapted to accented English with very little training. Training less than 1% of the parameters removed about 40% of the zero-shot word errors, and most of the gain came from the first few hundred utterances. This matters for care homes, because it suggests that a modest amount of local caregiver speech could adapt the model further without large infrastructure. The ablations also showed diminishing returns: doubling the rank from 16 to 32 gave a gain close to the run-to-run variation. A careful reading of the WER figure is still needed. WER was computed on raw transcripts without text normalisation, so part of the improvement may come from the adapted model learning the transcription style of the Svarah references (casing, punctuation and number formats) rather than only the accent.

Disfluency and self-corrections. The comparison of T5-small and BART-base shows why aggregate metrics are not enough for this task. On BLEU, ROUGE-L and exact match, the two models were close, and T5-small was slightly ahead. Yet on the cases that matter most in a care record, a corrected dose or a corrected time, BART-base was correct more often. A plausible explanation is BART's denoising pre-training, which teaches it to reconstruct a clean sentence from a corrupted one, a task that is close to resolving a repair. Neither model reliably resolved food self-corrections, and repairs with an unfamiliar editing phrase sometimes survived. These remaining errors are a reason to keep the original audio linked to each record.

Extraction and alerts. T5-small learned the 19-field format very well, but the difference between 100% alert recall on the original test split and 82.0% on the hand-written safety set is one of the most important results of this research. The generated dataset is regular, and the test split comes from the same process, so it rewards learning the dataset's phrasing. The safety set, with its different phrasings, gives a more realistic estimate. The error analysis then showed something useful: most alert errors were not failures to understand the observation but inconsistencies between the alert flag and the model's own fields. The model knew that "she had chest pain" was a symptom and that the symptom was chest pain, but still output "No". Treating the alert as a

consequence of the extracted fields, rather than as a separate prediction, fixed most of these errors. This is a design lesson that applies beyond this project: when a safety decision can be derived from structured fields with a clear rule, it is safer to derive it than to ask the model to predict it independently.

Long recordings. The failure of the PP2 pipeline on a longer recording was not caused by a weak model but by a mismatch between how the models were trained and how they were used. Whisper sees 30 seconds at a time, and T5-small was trained on single observations. Once the input was shaped to match the training conditions, by chunking the audio and splitting the sentences, every observation was recovered without retraining any model. The real recordings showed that the remaining errors in the complete pipeline are mostly upstream, in speech recognition of drug names and in sentence boundaries, which points to where future improvement should focus. This agrees with evidence from clinical dictation that a proportion of speech recognition errors in health records are clinically significant and need to be caught before the record is final [17].

7.2 Comparison With Literature Baselines and Existing Solutions

Javed et al. showed with the Svarah benchmark that existing English speech recognition systems struggle with Indian accents [6]. This research confirms that pattern with Whisper-small, which had a 26.13% WER on held-out Svarah utterances, and shows that LoRA adaptation can reduce it substantially. Because the test set used here is a random split of the same corpus, with speaker overlap, the result is not directly comparable with the original benchmark results or with other fine-tuning studies that use different data and recording conditions, and no claim of superiority over them is made.

Compared with the ADL extraction systems reviewed by Wieland-Jorna et al. [7], SCRIBE differs in its input and in how it addresses the data problem. The reviewed systems extract ADL information from written EHR notes, and data access was one of their main limitations. SCRIBE extracts from spoken observations and addresses data scarcity with a de-identified dataset grounded in real care-home records. Its weakness is the reverse of theirs: the data is accessible but synthetic, so its regularity overstates

performance on real language, as the safety evaluation showed. The approach of using an LLM to generate training data and then fine-tuning a small local model is similar to the one proposed by Tang et al. for clinical text mining [21], and the gap between the test split and the safety set in this research shows why such generated data should be complemented by independently written test cases.

Compared with ambient digital scribes [8], SCRIBE is narrower but more structured. Digital scribes produce narrative notes for clinicians and carry a risk of unsupported generated content that requires review. SCRIBE produces structured fields and fixed-template reports, so it cannot add content that was not extracted, and it adds an explicit alert decision that digital scribes do not provide. Compared with general disfluency correction work [9], [10], SCRIBE adds a domain-specific, safety-focused evaluation of self-corrections, which the general corpora do not test. Compared with rule-based filler removal, the fine-tuned models raised exact match from 25.6% to 64–67% and, importantly, resolved repairs that deleting fillers cannot resolve.

7.3 Research Contribution

The research contributions of this work are:

- A systematic accuracy-against-efficiency study of LoRA adaptation of Whisper-small for Indian-accented English, varying both training-data size and rank, which shows where returns diminish for a lightweight model.
- A controlled comparison of T5-small and BART-base repeated across two tasks, showing that the better architecture depends on the task: BART-base for resolving self-corrections and T5-small for structured 19-field extraction.
- A safety-weighted model selection approach, in which correct handling of clinically significant cases and alert recall are weighed above aggregate scores.
- Evidence that a held-out split from a generated dataset overstates alert reliability (100% against 82.0%), together with an error analysis showing that most alert errors were inconsistencies between the alert flag and the extracted fields, and that deriving the alert from the fields reduced missed alerts from 27 to 10.

- A de-identified ADL dataset of 2,325 records with a 19-field schema, grounded in real care-home records and caregiver input, and a 200-case safety-focused test set with caregiver-reviewed labels.

7.4 Product Contribution

The product contribution is a working documentation component for the CareSense platform. Caregivers can record observations through a web application that stays open on a shared computer, and SCRIBE turns each recording into structured ADL records, alerts for the supervisor, a written report and a spoken handover. The pipeline handles recordings longer than 30 seconds and recordings with several observations, protects medication and food names during grammar correction, and stores records in the shared CareSense database. It processes a 51-second recording in about 12 seconds on a GPU. The three trained models are small open models that run in the project's own backend, so recordings are not sent to an external AI model provider for processing; only the text-to-speech step currently uses an external web service (gTTS).

7.5 Limitations and Threats to Validity

Internal validity. Rank and data size for speech recognition were compared on the test set rather than the validation set, and each configuration was trained once, with run-to-run differences of about 0.1 to 0.2 WER points. WER was computed without text normalisation, and the zero-shot baseline used the openai-whisper transcription function while the adapted models used the Hugging Face generation function, so part of the measured gain may reflect transcription style. The BART-base and T5-small comparisons used different learning rates (and, for extraction, different numbers of epochs and decoding methods), so the extraction gap cannot be attributed to architecture alone. The choice of BART-base for disfluency removal rests on about 28 hand-written sentences rather than a large test set. The alert rules were designed after analysing errors on the 200-case safety set and were tested on the same cases, so the 93.3% result is optimistic.

External validity. The speech model was adapted on Indian-accented read and conversational speech, not on Sri Lankan caregivers, and the Svarah split was made by utterance, so test speakers also appear in training and the WER may be optimistic

for new speakers. The ADL dataset was generated by a language model from one facility's information, so its language is more regular than real caregiver speech, which is reflected in the gap between the test split and the safety set. The safety set itself was constructed for this work and uses the same 22 symptom types as training, so it tests new phrasings rather than new symptoms. The real-recording tests used a small number of recordings spoken by the author, not by caregivers, so they do not show how the system performs on the speech of different caregivers in the care home.

Construct validity and practical limitations. Event recall in the multi-observation experiment counts a category match, not the correctness of every field. The pipeline assumes one observation per sentence, so two events in one sentence ("I gave her the tablet and she ate breakfast") produce one record. Audio is cut at fixed positions, which can split a sentence, as happened in Run 2. Speech recognition errors on drug names are not protected by the vocabulary list. The proposal's context-aware module (using patient history and time of day) and explicit emotional observations were only partly realised: the time of day is extracted as a field, and emotional signs such as sudden crying appear only as symptoms. Processing on a single-thread CPU is slower than real time, the gTTS library relies on a web service whose terms may not permit commercial use, and the ethics decision for the research was still pending at the time of writing.

7.6 Implications

For care homes, the results suggest that voice-based documentation is feasible with small models and modest hardware, and that it can produce records that are more structured than handwritten notes. The results also show that such a system should keep a human in the loop: records should be reviewable against the original audio, and alerts should support, not replace, the supervisor. For researchers, the work shows the value of evaluating safety-relevant behaviour separately from aggregate metrics, of testing on phrasings that differ from generated training data, and of deriving safety decisions from structured fields where clear rules exist. For system designers, the long-recording problem shows that many failures in AI pipelines come from mismatches between training conditions and real use, and that reshaping the input can be more effective than retraining the model.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

SCRIBE's value proposition is simple: caregivers speak instead of writing, and the care home receives clear, structured records and timely alerts. For care homes that still rely on handwritten files, such as the ones visited in this research, this addresses three problems observed in the field: the time spent writing, the inconsistent format of handwritten notes, and observations that are missed or not passed on at handover. Table 8.1 summarises the target users and the value for each.

Table 8.1: Target users and value proposition

User group	Need	Value provided by SCRIBE
Caregivers	Record care quickly without leaving the care recipient or using a phone frequently	Voice recording on a shared web application; no handwriting
Supervisors and nurses	Know quickly when something needs attention; clear handover	Structured records, alerts on the dashboard, written and spoken handover
Care home management	Complete, consistent and searchable documentation	Digital records in a structured format that can be reviewed and reported
Care recipients and families	Safe, continuous care	Fewer missed observations and earlier attention to symptoms

The primary market is elder care homes in Sri Lanka that use paper records. Secondary markets could include nursing homes and home-care services where caregivers document routine activities, once the system has been validated with real caregiver speech.

8.2 Market and Competitive Position

SCRIBE competes with three kinds of existing solutions: handwritten records, typed electronic record systems, and AI documentation tools such as ambient digital scribes. Handwritten records are cheap but have the problems described above. Typed record systems give structured data but keep the documentation burden [4]. Digital scribes reduce documentation time for clinicians [8], but they are designed for doctor consultations, produce narrative notes and do not provide ADL fields or caregiver alerts.

SCRIBE's competitive position comes from its focus on the care-home setting. It is designed for ADL observations rather than consultations, it is adapted to South Asian-accented English and local medication and food names, it produces structured fields and fixed-template reports instead of free text, it includes an explicit alert decision, and it uses small models that can run on modest hardware. A detailed market size and pricing analysis was outside the scope of this individual component and should be carried out with the group before commercial launch.

8.3 Business or Sustainability Model

Several sustainability models are possible, and the final choice should be made at the CareSense level. A subscription per care home, scaled by the number of care recipients, would cover hosting and maintenance. A tiered model could offer basic recording and reports at a low cost, with alerts, dashboard analytics and integration with the other CareSense components in a higher tier. Partnerships with care-home groups, non-governmental organisations or public elder-care programmes could support adoption in homes with limited budgets.

The main running costs are hosting the backend and models, database and storage services, and maintenance. The timing results matter here: a GPU server processed a 51-second recording in about 12 seconds, while a single-thread CPU took 77 seconds, so the hosting choice is a trade-off between cost and waiting time. Long-term sustainability also depends on continuing to improve the models with real caregiver speech, which requires a consent-based data collection process agreed with the participating care homes.

8.4 Scalability, Intellectual Property and Compliance

Scalability. Because the pipeline is modular, each stage can be improved or replaced independently, and a care home is added by adding its care recipients and caregivers to the shared database. Processing capacity scales with the backend hardware. The protected vocabulary and alert rules are simple lists and rules that can be extended for each facility.

Intellectual property. The project's own intellectual property includes the trained LoRA adapter and fine-tuned models, the 19-field ADL schema, the de-identified

ADL dataset, the safety-focused test set, the alert rules and the pipeline design. The base models and libraries are open source and remain subject to their own licences, and the public training datasets (Svarah, DisfluencySpeech and DISCO) have their own terms of use. All of these licences must be reviewed for commercial use. The gTTS library uses a Google web service, and a commercial deployment would likely need a licensed text-to-speech service instead.

Compliance. Sri Lanka's Personal Data Protection Act, No. 9 of 2022, classes data concerning health as a special category of personal data [28]. A commercial deployment must therefore obtain consent, limit access to authorised staff, store records and audio securely, define retention periods and allow care recipients' data rights to be exercised. Ethics approval and a formal agreement with each care home are required before real care recipient data or caregiver voice recordings are used to improve the models.

8.5 Contribution to the Sustainable Development Goals

The 2030 Agenda for Sustainable Development sets out 17 Sustainable Development Goals (SDGs) [40]. SCRIBE contributes mainly to SDG 3, as proposed in the original research proposal, and also supports three further goals (Table 8.2).

Table 8.2: Contribution of SCRIBE to the Sustainable Development Goals

SDG	Contribution
SDG 3: Good Health and Well-Being	Better documentation of the daily care of older adults, earlier attention to symptoms and abnormal vital signs, and clearer handover between caregivers.
SDG 8: Decent Work and Economic Growth	Less time spent on paperwork, reducing caregiver workload and stress and leaving more time for direct care.
SDG 9: Industry, Innovation and Infrastructure	Brings AI-based documentation to a care sector that still relies on paper records.
SDG 10: Reduced Inequalities	Speech recognition adapted to South Asian accents and local vocabulary rather than only to native English speakers.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This research set out to help caregivers in care homes document Activities of Daily Living by voice instead of by hand. Field visits to the Love & Elder Care Home showed that handwritten records are hard to maintain and that observations are sometimes missed. SCRIBE was developed as the documentation component of the CareSense framework to address this problem. It converts a caregiver's voice log into structured, 19-field ADL records, decides whether each observation needs the supervisor's attention, and produces a written report and a spoken handover.

The results show that each stage can be made to work with small, open models. LoRA adaptation reduced the speech recognition error rate on accented English from 26.13% to 15.57% while training less than 1% of the model. A fine-tuned BART-base model resolved spoken self-corrections, including medication doses and times, better than the alternatives, and a fine-tuned T5-small model extracted structured records with 92.3% exact match on its test split. The research also showed the limits of evaluating on generated data: alert recall fell from 100% on the test split to 82.0% on a separate safety set. Most of these missed alerts were inconsistencies between the alert flag and the extracted fields, and deriving the alert from the fields raised recall to 93.3% with no false alarms. Finally, audio chunking and sentence segmentation allowed the system to process long recordings with several observations, and a 51-second recording was processed in about 12 seconds on a GPU.

SCRIBE is therefore a working and evaluated prototype, not yet a validated clinical tool. Its main remaining weaknesses are speech recognition of drug names, the regularity of the generated training data, and the lack of an on-site evaluation with real caregiver speech, which is the most important next step.

9.2 Achievement of Objectives

Table 9.1 summarises how far each specific objective was achieved.

Table 9.1: Achievement of the specific objectives

Objective	Status	Evidence
1. Examine current ADL documentation methods and their challenges	Achieved	Field visits, handwritten records and caregiver conversations at the care home; the same problem confirmed at a second care home (Sections 1.2 and 3.2.1).
2. Speech recognition for caregiver voice logs	Achieved, with limitations	WER reduced from 26.13% to 15.57% on accented English; long recordings supported. Not yet adapted to real caregiver speech; drug names remain error-prone (Sections 6.1.1 and 6.2).
3. Auto-editing of fillers, repetitions and self-corrections	Achieved, with limitations	Exact match raised from 25.6% to 64.0%; BART-base resolved safety-relevant dose and time corrections; some food corrections unresolved (Section 6.1.2).
4. Domain-specific extraction of ADL information	Mostly achieved	19-field extraction with 92.3% exact match; alert recall 82.0% on the safety set and 93.3% with rules. Emotional observations are covered only as symptoms, and the context-aware use of patient history was not implemented (Sections 6.1.3 to 6.1.5).
5. Structured records, handover summaries (text and audio) and evaluation	achieved	Records, written reports and spoken handovers implemented and evaluated for accuracy and completeness; usability and impact on continuity of care not yet measured in the care home (Sections 4.4.8, 6.2 and 6.3).

9.3 Individual Research and Product Contributions

The author's individual research contributions are the LoRA data-efficiency and rank study for accented speech recognition, the repeated T5-small and BART-base comparison with safety-weighted selection, the de-identified ADL dataset and the 200-case safety test set, and the finding that alert errors can be reduced by deriving the alert from extracted fields. The individual product contributions are the SCRIBE pipeline with audio chunking, sentence segmentation, protected-vocabulary grammar correction, 19-field extraction, alert rules, template reports and spoken handover, together with the SCRIBE web application and backend connected to the shared CareSense data layer.

9.4 Future Directions

- **On-site evaluation and real caregiver speech.** Test SCRIBE in the care home with caregivers recording real observations, once ethics approval is granted, and measure accuracy, usability and time saved. With consent, use this speech to adapt the speech recognition model to Sri Lankan caregivers.

- **Hands-free recording.** Add a wearable microphone so that caregivers can record observations while giving care, without using a phone or a computer.
- **Multi-observation training.** Train the extraction model on inputs that contain several observations, built within each data split to avoid leakage, so that two events in one sentence are both extracted.
- **Better segmentation of audio.** Cut audio at pauses rather than at fixed positions, so that sentences are not split between pieces.
- **Protecting drug names in speech recognition.** Add the medication list to the recognition stage, for example through vocabulary biasing or correction against the facility's medication list, so that names such as Losartan are recognised correctly before grammar correction.
- **Stronger alert logic.** Add numeric thresholds for vital signs (for example blood sugar or blood pressure values), validate the alert rules on a new test set, and extend the schema to emotional observations and the context-aware use of patient history and time of day proposed originally.
- **Efficiency and languages.** Reduce CPU processing time through model quantisation or faster inference engines, and extend the system to Sinhala, Tamil and code-mixed speech.

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GLOSSARY

Activities of Daily Living - Basic self-care tasks such as eating, bathing, dressing, moving and taking medication, used to describe a person's daily functioning and care needs.

Alert recall - The share of observations that genuinely need an alert which the system actually flagged.

Beam search - A decoding method that keeps several candidate outputs at each step and returns the most probable complete output.

De-identification - Removing or replacing information, such as names and ages, that could identify a person.

Disfluency - An interruption in the flow of speech, such as a filler ("uh"), a repetition or a self-correction.

Exact match - An evaluation measure that counts an output as correct only if it is identical to the reference.

Fine-tuning - Further training of a pre-trained model on task-specific data.

Greedy decoding - A decoding method that chooses the most probable token at each step.

Handover - The transfer of information about care recipients from one caregiver or shift to the next.

Low-Rank Adaptation - A parameter-efficient fine-tuning method that freezes a pre-trained model and trains small low-rank matrices added to selected layers.

Object storage - A storage service that keeps files, such as audio clips, as objects identified by keys.

Self-correction (repair) - A part of speech in which the speaker replaces something already said, for example "two, no wait, one tablet".

Sentence segmentation - Splitting a transcript into separate sentences so that each can be processed on its own.

Word error rate - The number of word substitutions, deletions and insertions needed to turn a transcript into the reference, divided by the number of reference words.

Zero-shot - Using a pre-trained model on a task without any task-specific training.

APPENDICES

Appendix A Extended Results and Research Materials

A.1 Symptom-level results on the safety set

Table A.1 shows how many of the five phrasings of each symptom type were flagged by T5-small without the alert rules. All 22 symptom types are alert-worthy according to the care home's alert definition.

Table A.1: Symptom-level alert recall of T5-small on the safety set (five phrasings each)

Symptom	Detected	Missed	Recall (%)
body pain	5	0	100
breathing difficulty	5	0	100
chest pain	5	0	100
difficulty passing urine	5	0	100
fever	5	0	100
headache	5	0	100
inflammation	5	0	100
bed sore	4	1	80
extreme fatigue	4	1	80
fainting	4	1	80
hand pain	4	1	80
leg pain	4	1	80
loss of consciousness	4	1	80
skin allergy	4	1	80
stomach pain	4	1	80
unusual fatigue	4	1	80
frequent urination	3	2	60
loose motion	3	2	60
speaking unrelated words	3	2	60
sudden crying	3	2	60
mild coughing	2	3	40
vomiting	2	3	40

A.2 Examples of safety-set sentences

Table A.2: Example sentences from the safety-focused test set

Group	Example sentence	Expected alert
Symptom (bed sore)	P01 was noticed to have a bed sore this morning.	Yes
Symptom (vomiting)	P19 vomited this morning.	Yes
Meal (partial)	P02 ate only half of the bread served for breakfast.	Yes
Meal (refused)	P14 refused all of the food offered at breakfast.	Yes
Vital sign	Nurse checked P01's blood pressure, reading was 155/90 and was noted as elevated.	Yes
Control (normal vital)	P14 had a blood pressure reading of 115/65 and it was recorded as normal.	No
Control (hygiene)	P01 received a sponge bath this morning.	No

A.3 Examples of domain-specific disfluency pairs

Table A.3: Examples of domain-specific disfluency training pairs

Disfluent input	Clean target
She took Atova, um, 10mg after breakfast.	She took Atova 10mg after breakfast.
Glyco was given, 80mg, uh, before lunch today.	Glyco 80mg was given before lunch today.

A.4 Example of an ADL training target

Sentence: "P23 complained of headache after lunch."

Target string: "category: symptom | medication: | dosage: | food: | meal_type: | intake: | fluid: | amount: | hygiene: | mobility: | destination: | symptom: headache | vital: | reading: | status: | visitor: | reason: | time: evening | alert: Yes"

Appendix B Academic Integrity and AI Use Disclosure

Generative AI tools were used in this research only as support, in the ways listed in Table B.1. The research itself, including the field study, the de-identification and checking of the data, the implementation, the model training, the testing, the analysis and the results, was carried out by the author. The author remains responsible for all data, code, analyses, citations and conclusions in this thesis.

Table B.1: Use of generative AI tools

Purpose	How the tool was used	Work carried out by the author
Generating the ADL dataset (Claude [23])	Generated caregiver-style observation sentences, their 19 fields and alert labels from de-identified information typed by the author	Collected and typed the care-home information, removed names and ages, assigned P-codes, defined the alert rules with caregivers, and checked every generated record against the handwritten records and the caregivers' information (ADL dataset)
Coding support	Guidance and suggested code for implementation and evaluation steps, used commonly across the project	Designed and ran all experiments in Colab, tested and corrected the code, and checked every output; all reported numbers come from the author's runs
Writing support	Assisted in following the required template structure, provided guidance on drafting text, and helped improve language, grammar and clarity across the project documentation and the thesis	Provided all research content, results and information, and carefully reviewed, edited and approved the final text, figures and references in both the project documentation and the thesis